SHRI VILE PARLE KELAVANI MANDAL's

USHA PRAVIN GANDHI COLLEGE OF ARTS, SCIENCE AND COMMERCE

Bhakti Vedanta Swami Marg, North-South Road No. 1

Juhu Scheme, Vile Parle (West), Mumbai 400 056.

Accreditation by NAAC A+ Grade with CGPA 3.27

(AUTONOMOUS)

Affiliated to the

UNIVERSITY OF MUMBAI

Program: Bachelor of Science
B. Sc. (Information Technology)

Semester V & VI

Choice Based Credit System (CBCS)
under NEP 2020
with effect from the Academic year 2026 - 27

Academic Council No:

Agenda No:

Preamble

The Bachelor of Science in Information Technology (B.Sc-IT) program, is aligned with the visionary National Education Policy (NEP) 2020, which embarks on a holistic and interdisciplinary educational journey. This program is designed to meet the dynamic needs of the digital age, ensuring our graduates are not only proficient in the technical domain but also emerge as adaptable, creative, and ethical contributors to society.

At the core of this program is a curriculum that integrates technical mastery with a broad spectrum of knowledge, embracing the complexity and interconnectivity of today's global challenges. Through a blend of Major subjects, Minor specializations, Open Electives, Vocational and Skill Enhancement Courses, and comprehensive Ability Enhancement Courses, the program offers a rich, diversified learning experience. This approach ensures that students are well-versed in the latest technological advancements while being deeply rooted in ethical values and cross-disciplinary knowledge.

Third Year Overview

Semester V and VI represent the culmination of the B.Sc. Information Technology program, where students transition from foundational and intermediate concepts to advanced technologies and industry-oriented practices. These semesters emphasize specialization, innovation, and professional readiness, preparing students for careers in modern computing environments.

Semester V focuses on strengthening enterprise-level development skills and emerging technologies. Core Major subjects such as Enterprise Java, Artificial Intelligence, Security in Computing, Design Principles of User Interface and Experience, and Software Project Management equip students with the knowledge required to design, develop, and manage complex software systems. Accompanying laboratory courses such as Enterprise Java Lab, Artificial Intelligence Lab and Security in Computing Lab provide practical exposure to real-world application development, system security, and modern development frameworks.

The Minor component introduces interdisciplinary technologies like Internet of Things, enabling students to explore automation and smart system integration. Vocational and Skill Enhancement Courses such as Android Programming Lab ensure that students gain essential industry skills related to mobile application development and collaborative software development practices. The semester also includes a Field Project, giving students an opportunity to apply theoretical knowledge to practical problem-solving in real-world scenarios.

Semester VI further advances students' expertise with cutting-edge domains such as Data Science, Cloud Computing, DevOps, Advanced Artificial Intelligence, and Ethical Hacking. These courses focus on modern computational techniques, large-scale data analysis, cloud-based infrastructure, and cybersecurity practices, all of which are highly demanded in the IT industry. Practical laboratories complement these subjects by allowing students to work with real datasets, cloud platforms, and security tools to strengthen their technical competence.

The Minor component in this semester introduces Search Engine Optimization, expanding students' understanding of digital presence and web visibility, while practical exposure is enhanced through Hybrid Mobile Application Development and a Project module. A key highlight of the semester is On-Job Training (OJT), where students gain valuable industry experience by working in professional environments. This training enables them to understand real organizational workflows, apply technical knowledge in live projects, and develop professional skills such as teamwork, communication, and problem-solving.

Across both semesters, the third year emphasizes advanced technical mastery, industry exposure, and professional development. Through project-based learning, practical laboratories, and industry training opportunities, students graduate with a comprehensive understanding of modern IT technologies and the confidence to contribute effectively to the rapidly evolving technology landscape.

The B.Sc-IT program under NEP 2020 offers a transformative educational experience that holistically prepares students for the challenges and opportunities of the future. By balancing technical education with ethical, environmental, and cross-disciplinary learning, the program aims to cultivate professionals who are not only skilled in information technology but are also equipped to lead with innovation, integrity, and responsibility in the global digital landscape.

PROGRAMME SPECIFIC OUTCOMES (PSO'S)

Upon completing the B.Sc. in Information Technology, learners will be able to:

PSO1: Master theoretical concepts and apply practical IT skills to solve complex problems in real-world scenarios.

PSO2: Demonstrate analytical skills by designing and evaluating IT solutions with modern tools and data-driven approaches.

PSO3: Acquire fieldwork experience, developing essential soft skills for leadership and teamwork within the IT sector.

PSO4: Uphold ethical standards and sustainable practices in IT, exemplifying responsible professional behavior.

PSO5: Embrace lifelong learning, continuously adapting to emerging technologies and trends to stay relevant in the industry.

Pedagogy

At SVKM's Usha Pravin Gandhi College of Arts, Science, and Commerce (Autonomous), the B.Sc. IT program's pedagogical framework is designed to offer students a rich and dynamic educational experience, blending traditional teaching methodologies with cutting-edge technological tools and innovative learning strategies. This approach ensures that students are not only well-versed in fundamental IT principles but are also adept at applying this knowledge in a rapidly evolving technological landscape.

Enhanced Teaching Learning Methods for B.Sc. IT:

- **Classroom Lectures with Smartboards:** Utilize smartboards to make lectures more interactive, facilitating a deeper understanding of IT concepts through dynamic content display and engagement.
- **Experiential Learning:** Enable students to apply theoretical knowledge to real-world IT challenges through hands-on projects and simulations, bridging the gap between theory and practice.
- **Team-Based Learning:** Encourage collaborative learning through group IT projects, fostering teamwork, leadership, and personalized learning experiences.
- **Flipped Classroom:** Leverage online resources for pre-class study, allowing in-class time to focus on discussions and practical applications of IT concepts.
- **Group Discussion:** Promote the exchange of ideas and critical thinking on emerging IT trends and issues, enhancing communication and analytical skills.

Innovative and Technological Methods Tailored for B.Sc. IT:

- **Technology Integration in Learning:** Integrate immersive digital tools like Padlet, Quizlet, Kahoot, and Mindmeister in coursework, making complex IT subjects accessible and engaging.
- **Peer Teaching:** Encourage student-led instruction, where peers share knowledge and insights, fostering a deeper understanding of IT subjects and leadership qualities.
- **Blended Learning Approach:** Combine the benefits of smartboard-enhanced classroom lectures with extensive online resources, offering a flexible and comprehensive learning experience.
- **Community-Based Learning:** Apply IT skills to community service projects, reinforcing the importance of social responsibility and ethical technology use.

By incorporating smartboards in classrooms, we enhance the visual and interactive aspects of teaching, making complex subjects more engaging and understandable. Online resources complement traditional methods, providing students with the flexibility to learn at their own pace and deepen their understanding of information technology.

The college's commitment to innovative teaching and the use of advanced technology in education ensures that our students are not only academically proficient but also equipped with the practical skills and ethical understanding necessary to navigate the challenges of the IT industry. This holistic approach prepares our graduates for excellence in their academic endeavors and their future careers, setting them on the path to becoming leaders in the field of information technology.

Signature
HOD

Signature
Approved by I/C Principal

Credit Distribution Structure for

Third Year (B.Sc. (Information Technology))

SEM V				
Sr. No.	**Name of the Module (Subject)**	**Module Code**	**Module Category** **(Core, Core Elective, OE,VSC, SEC, AEC,VSC, IKS, CC, FP, OJT, RM, CEP, RP)**	**Total no. of credits**
1	Computational Indic Studies	UICIS301	MAJ	**2**
2	Enterprise Java	UIENJ302	MAJ	**2**
3	Design Principles Of User Interface And Experience	UIPUI303	MAJ	**2**
4	Design Principles Of User Interface And Experience LAB	UIFSD303P	MAJ	**1**
5	Software Project Management	UISPM304	MAJ	**2**
6	Software Project Management LAB	UISPM304P	MAJ	**1**
7	Artificial Intelligence	UIARI305	MAJ	**3**
8	Artificial Intelligence LAB	UIARI305P	MAJ	**1**
9	Security in Computing	UISIC306	MAJ	**3**
10	Security in Computing LAB	UISIC306P	MAJ	**1**
11	Internet Of Things	UIIOT307	MIN	**3**
12	Internet Of Things LAB	UIIOT307P	MIN	**1**
13	Android Programming LAB	UIANP308P	VSC/SEC	**1**
14	Enterprise Java LAB	UIENJ302P	VSC/SEC	**1**
15	Field Project	UIFPR310	FP	**2**
16	Community Engagement Project	UICEP311	CEP	**2**

SEM VI				
Sr. No.	**Name of the Module (Subject)**	**Module Code**	**Module Category (Core, Core Elective, OE,VSC, SEC, AEC,VSC, IKS, CC, FP, OJT, RM, CEP, RP)**	**Total no. of credits**
1	Introduction to Devop	UIITD351	MAJ	**2**
2	Introduction to Devop LAB	UIITD351P	MAJ	**1**
3	Data Science	UIDAS352	MAJ	**3**
4	Data Science LAB	UIDAS352P	MAJ	**1**
5	Cloud Computing	UICLC353	MAJ	**2**
6	Cloud Computing LAB	UICLC353P	MAJ	**1**
7	Advanced Artificial Intelligence	UIAAI354	MAJ	**3**
8	Advanced Artificial Intelligence LAB	UIAAI354P	MAJ	**1**
9	Ethical Hacking	UIETH355	MAJ	**3**
10	Ethical Hacking LAB	UIETH355P	MAJ	**1**
11	Search Engine Optimization	UISEO356	MIN	**2**
12	Project	UIPRJ357	MIN	**1**
13	Hybrid Mobile Application LAB	UIHMA358 P	MIN	**1**
14	On Job Training	UIOJT359	OJT	**4**

Sign of HOD
Dr. Smruti Nanavaty
Dept of Information Technology

Sign of I/C Principal
Dr. Smruti Nanavaty

Syllabus

B.Sc. (Information Technology)

(Semester V & VI)

SEMESTER V

<table>
<tr><td colspan="3">Programme : B. Sc. Information Technology (2026 - 27)</td><td>Semester V</td></tr>
<tr><td colspan="3">Course : Computational Indic Studies</td><td>Code : UICCIS301</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td rowspan="2">Lecture</td><td rowspan="2">Credits</td><td colspan="2"></td></tr>
<tr><td>Continuous Assessment (CA) (Marks - 20)</td><td>Semester End Examinations (SEE) (Marks- 30)</td></tr>
<tr><td>02 Lab based</td><td>2</td><td>20</td><td>30</td></tr>
<tr><td colspan="4"></td></tr>
<tr><td colspan="4">Internal Component (Practical Break up)</td></tr>
<tr><td colspan="2">Examination</td><td colspan="2">Mini Project/Case study/ Field Visit (report to be submitted and certified prior to practical examination)</td></tr>
<tr><td colspan="2">30 Marks</td><td colspan="2">20 Marks</td></tr>
</table>

Learning Objectives

CO1: To understand the evolution, structure, and linguistic features of major historical scripts used in Indian knowledge traditions.

CO2: To analyze challenges in digitizing historical manuscripts and apply basic computational tools such as Unicode encoding and OCR for Indic scripts.

CO3: To evaluate the role of Artificial Intelligence and Natural Language Processing in manuscript analysis, restoration, and script recognition.

CO4: To explain the importance of digital technologies in preserving Indian knowledge systems and cultural heritage.

CO5: To apply immersive technologies such as Augmented Reality (AR) and Virtual Reality (VR) to conceptualize digital heritage platforms.

CO6: To design a prototype concept for an IKS Digital Museum integrating AI, AR, and VR for knowledge preservation and interactive learning.

Learning Outcomes:

After successful completion of the course, students will be able to:

1. Describe the evolution and structural features of major scripts used in the Indian subcontinent.
2. Explain key characteristics of Sanskrit, Pali, Arabic, and Gurmukhi manuscript traditions.
3. Analyze challenges involved in digitizing historical manuscripts including script variation and document degradation.
4. Apply basic digital tools such as Unicode encoding and OCR for processing Indic scripts.
5. Explain the role of Artificial Intelligence, Augmented Reality, and Virtual Reality in preserving Indian knowledge systems.
6. Design a conceptual prototype for an IKS Digital Museum using immersive technologies.

Pedagogy :

Lab based

<table>
<tr><th>Module</th><th>Module Content</th><th>Duration of Module</th></tr>
<tr><td>I</td><td>Scriptology and Digital Manuscript Analysis (Lab-Oriented)

Introduction to Scriptology and the evolution of writing systems in the Indian subcontinent.

Study of major classical and medieval scripts including Sanskrit script traditions, Pali manuscript traditions, Arabic script in medieval scholarship, and Gurmukhi script.

Structural characteristics of scripts including phonetic representation, writing patterns, and orthographic systems.

Digitization challenges in historical manuscripts including script variations, document degradation, and multilingual complexity.

Practical introduction to Unicode encoding, OCR technologies, and digital text processing for Indic scripts.

Applications of Artificial Intelligence and Natural Language Processing for script recognition, manuscript restoration, and textual analysis.</td><td>15</td></tr>
<tr><td>II</td><td>Immersive Technologies for Preserving Indian Knowledge
Digital heritage preservation using modern computing technologies.

Introduction to Augmented Reality (AR) and Virtual Reality (VR) for cultural and knowledge preservation.

Concept of IKS Virtual Museums:
Digitally recreating ancient libraries, manuscripts, and knowledge systems using immersive technologies.

Applications such as:
• VR reconstruction of ancient universities and knowledge centers
• AR-based interactive learning of scripts and manuscripts
• 3D visualization of ancient instruments and knowledge artifacts

Use of AI, AR, and VR to build interactive knowledge repositories.

Case studies of digital heritage initiatives and computational platforms used to preserve cultural knowledge.

Design concept: IKS Digital Museum Prototype where students conceptualize immersive exhibits of Indian knowledge traditions.</td><td>15</td></tr>
</table>

Books and References:

SrNo	Dataset/Books	Authors	Publisher	Date
1	Datasets Indian Scripts		Kaggle	
2	Word Level Script Identification Using Convolutional Neural Network Enhancement for Scenic Images ACM Transactions on Asian and Low-Resource Language Information Processing 21(4):1-29 DOI:10.1145/3506699			March 2022
3	Synthetically Created Dataset Historic Scripts to Modern Vision: A Novel Dataset and A VLM Framework for Transliteration of Modi Script to Devanagari		https://aikosh.indiaai.gov.in/home/datasets/details/modescript_synthetic_dataset.html	2025
4	Introduction to Indian Knowledge System: Concepts and Applications	B. Mahadevan, Vinayak Rajat Bhat & Nagendra Pavithran	PHI Learning Covers scientific traditions, mathematics, knowledge organization, and modern applications of IKS.	2022

<table>
<tr><td colspan="2">Program: B.Sc. – Information Technology</td><td colspan="2">Semester: V</td></tr>
<tr><td colspan="2">Course: Enterprise Java (2026-2027)</td><td colspan="2">Course Code: UUGADJ57</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td>Lecture
(Hours per week)</td><td>Credit</td><td>Continuous Assessment (CA)
(Marks - 20)</td><td>Semester End Examinations (SEE)
(Marks- 30)</td></tr>
<tr><td>4</td><td>2</td><td>20</td><td>30</td></tr>
<tr><td colspan="4">Learning Objectives:
To understand Java EE: What constitutes an Enterprise Application and Java Enterprise Edition.
To learn about request dispatcher, cookies, session, and file handling
To get started with Java Server Pages and JSTL
To gain knowledge of working with Session Beans, Persistence, Object/Relational Mapping, JPA and Hibernate</td></tr>
<tr><td colspan="4">Course Outcomes:
After completion of the course, learners would be able to:
CO1: understand the concept of enterprise edition of java and its architecture
CO2: learn request dispatcher, cookies, session, and file handling
CO3: start developing JSP and JSTL based EE JAVA applications
CO4: learn the working with Session Beans, Persistence, Object/Relational Mapping, JPA and Hibernate</td></tr>
<tr><td colspan="4"></td></tr>
<tr><td colspan="4">Outline of Syllabus: (per session plan)</td></tr>
</table>

<table>
<tr><td>Modules</td><td>Topics</td><td>Duration (Lecture)</td></tr>
<tr><td>Module 1</td><td>Introduction to EE and Java Servlet</td><td>10</td></tr>
<tr><td></td><td>Understanding Java EE: What is an Enterprise Application? What is java enterprise edition? Java EE Technologies, Introduction to Java Servlets: The Need for Dynamic Content, Java Servlet Technology, Why Servlets? What can Servlets do? Servlet API and Lifecycle: Java Servlet API, The Servlet Skeleton, The Servlet Life Cycle, A Simple Welcome Servlet Working with Servlets: Getting Started, Using Annotations Instead of Deployment Descriptor
Request Dispatcher, Cookies, Session, File handling
Request Dispatcher: Resquestdispatcher Interface, Methods of Request dispatcher, Request dispatcher Application. COOKIES: Kinds Of Cookies, Where Cookies Are Used? Creating Cookies Using Servlet, Dynamically Changing the Colors of a Page SESSION: What Are Sessions? Lifecycle Of Http Session, Session Tracking with Servlet API, A Servlet Session Example Working with Files: Uploading Files, Creating an Upload File Application, Downloading Files, Creating a Download File Application.</td><td></td></tr>
<tr><td>Module 2</td><td>JSP and JSTL</td><td>10</td></tr>
<tr><td></td><td>Introduction To Java Server Pages: Why use Java Server Pages? Disadvantages Of JSP, JSP v\s Servlets, Life Cycle of a JSP Page, how does a JSP function? How does JSP execute? About Java Server Pages</td><td></td></tr>
</table>

	Getting Started with Java Server Pages: Comments, JSP Document, JSP Elements, JSP GUI Example. Action Elements: Including other Files, Forwarding JSP Page to Another Page, Passing Parameters for other Actions, Loading a JavaBean. Implicit Objects, Scope, and El Expressions: Implicit Objects, Character Quoting Conventions, Unified Expression Language [Unified El], Expression Language. **Java Server Pages Standard Tag Libraries:** What is wrong in using JSP Scriptlet Tags? How JSTL Fixes JSP Scriptlet's Shortcomings? Disadvantages Of JSTL, Tag Libraries.	
Module 3	**EJB, ORM, JPA**	**10**
	Introduction To Enterprise JavaBeans: Enterprise Bean Architecture, Benefits of Enterprise Bean, Types of Enterprise Bean, Accessing Enterprise Beans, Enterprise Bean Application, Packaging Enterprise Beans Working with Session Beans: When to use Session Beans? Types of Session Beans, Remote and Local Interfaces, Accessing Interfaces, Lifecycle of Enterprise Beans, Packaging Enterprise Beans, Example of Stateful Session Bean, Example of Stateless Session Bean, Example of Singleton Session Beans. **Persistence, Object/Relational Mapping And JPA:** What is Persistence? Persistence in Java, Current Persistence Standards in Java, why another Persistence Standards? Object/Relational Mapping, Introduction to Java Persistence API: The Java Persistence API, JPA, ORM, Database and the Application, Architecture of JPA, How JPA Works? JPA Specifications. Writing JPA Application: Application Requirement Specifications, Software Requirements,	
Total Lectures		**60**

Essential readings:

1. Java EE 7 For Beginners Sharanam Shah, Vaishali Shah, SPD, First edition, 2017
2. Java EE 8 Cookbook: Build reliable applications with the most robust and mature technology for enterprise development, Elder Moraes, Packt, First Edition, 2018
3. Advanced Java Programming, Uttam Kumar Roy, Oxford Press, 2015

Reference books:

1. "Java EE 7 Essentials", Arun Gupta, O'Reilly Media, 1st Edition, 2013
2. "Java Persistence with Hibernate", Christian Bauer, Gavin King, and Gary Gregory, Manning Publications, 1st Edition, 2007
3. "Pro EJB 3: Java Persistence API", Mike Keith, Merrick Schincariol, Apress, 1st Edition, 2006

Program: B.Sc. – Information Technology (2026 - 27)				Semester: V
Course: Design Principles of User Interface and Experience				Course Code: UIPUI303
Teaching Scheme		Evaluation Scheme		
Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks - 20)	Semester End Examinations (SEE) (Marks - 30 in Question Paper)	
2	2	20	30	

Learning Objectives:

1. Understand the foundational concepts of UI/UX design, differentiate between UI and UX, and explore user-centered design principles and design thinking methodologies
2. Apply cognitive and behavioral psychology principles, such as Gestalt, Jakob's Law, and Fitt's Law, in creating user-friendly and intuitive designs
3. Analyze and implement advanced practices in UI/UX design, including responsive and adaptive design, wireframing, and micro-interactions, while exploring emerging trends and ethical considerations

Course Outcomes:

After completion of the course, learners would be able to:

CO1: Remember the basic principles of good design, such as layout, color theory, and typography, and differentiate between UI and UX

CO2: Understand user-centered design concepts, empathy maps, and personas, and how psychological principles like Miller's Law and Hick's Law influence design decisions

CO3: Apply responsive design techniques and wireframing tools to create low-fidelity prototypes for applications or websites

CO4: Analyze the UI/UX of popular applications, such as Instagram or Google Maps, to evaluate usability and navigation flow

CO5: Evaluate the accessibility of applications using tools and guidelines, such as WCAG, and critique poorly designed interfaces for improvement opportunities

CO6: Create innovative and user-centric design solutions, including wireframes and prototypes, while integrating principles of cognitive psychology and emerging trends like Lean UX and micro-interactions

Outline of Syllabus: (per session plan)

Module	Description	No of Hours
1	**Foundations of UI/UX Design** Definitions: UI vs. UX, Principles of good design, Introduction to user research: Personas, user journeys, and empathy maps, Overview of design thinking, User centred design	08

	Theory and Design Principles Grid, Color Theory, Typography and iconography, White space, Layout and hierarchy, Content, User experience, Images and imagery, Extra tidbits ***Case Study as individual or group activity*** 1. Analyze the navigation flow of any application 2. Analyze the UI/UX of popular apps like Google Maps, Instagram, or Duolingo for its usability 3. Analyzing the UX of a social media app (e.g., Netflix or Spotify) 4. Evaluating the success stories of minimalistic designs (e.g., Apple) and their impact on usability 5. Design critique of a poorly designed website or app Test the accessibility of an app (e.g., YouTube or Khan Academy) using accessibility tools and guidelines (like WCAG)	
2	**Cognitive and Behavioural Principles in UX Design** A Brief History of Psychology and Design Gestalt Psychology, Human Factors Engineering, Human–Computer Interaction, User Experience Design, Fidelity and Prototyping Applying Psychological Principles in Design Building Awareness, Design Principles, Jakob's Law, Fitt's Law, Miller's Law, Hick's Law, Postel's Law, Peak-End Rule, Aesthetic-Usability Effect, Von Restorff Effect, Tesler's Law ***Case Study as individual or group activity*** 1. How any application uses design thinking to solve user problems (e.g. Airbnb and Uber) Case-lets for psychological principles of design listed in the syllabus	12
3	**Advanced Practices and Emerging Trends in UX Design** Lean UX, Responsive and adaptive design principles, Wire-framing, Micro interactions, Tools for UX design, Future of UX, Ethics in UI/UX **Figma's AI Plugins (Magician, Diagram) – Auto-generates UI components, copy, and design structures** ***Case Study as individual or group activity*** 1. Examine the redesign process of a popular website or app (e.g., Instagram's dark mode) 2. Compare the mobile and desktop versions of a popular website (e.g., LinkedIn or Medium) to assess responsive design principles 3. Create a low-fidelity prototype based on mental models and cognitive psychology insights Creating wireframes for an E-commerce website	10
Total		**30**

Reference Books:

[1] “Introduction to Design Thinking for UX Beginners”, Uijun Park, 2023

[2] “Principles of UX The Pocket Universal Principles of UX”, Irene Pereyra, 2024

[3] “Hello Web Design”, Tracy Osborn, O’Reilly, 2021

[4] “Steal Like an Artist”, Austin Kleon, 2012

[5] “UX Mastery: The Art & Science of User Experience Design”, Mayur Chaudhary, Kishore Kankipati, 2024

Program: B.Sc. – Information Technology (2026 - 27)	Semester: V
Course: Design Principles Of User Interface And Experience LAB	Course Code: UIFSD303P

Teaching Scheme		Evaluation Scheme	
Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks - 20)	Semester End Examinations (SEE) (Marks - 30 in Question Paper)
2	1	20	30

Learning Objectives:

1. Understand foundational principles of full-stack web development, including front-end frameworks, back-end architecture, and API integration
2. Apply tools and techniques to build, deploy, and test full-stack web applications, leveraging modern frameworks and libraries
3. Analyze real-world requirements to design and implement scalable and secure full-stack solutions with effective communication between front-end and back-end

Course Outcomes:

After completion of the course, learners would be able to:

CO1: List the tools, libraries, and frameworks such as Vue.js, React, Express.js, and MongoDB, along with their key functionalities, used in full-stack web development

CO2: Explain how front-end frameworks like Vue.js or Angular interact with back-end APIs (REST/GraphQL) to enable data communication and dynamic web functionalities

CO3: Build a dynamic web application that includes front-end components like forms or modals, integrates state management, and connects to a back-end API for CRUD operations

CO4: Compare server-side rendering (SSR) and client-side rendering (CSR) techniques, evaluating their impact on performance, SEO, and user experience

CO5: Assess the scalability, performance, and security of a deployed full-stack web application, identifying areas for optimization and suggesting improvements

CO6: Design and develop a real-time, full-stack web application (e.g., chat application) that incorporates WebSocket communication, role-based authentication, and database integration

Outline of Syllabus:

	List of Practicals :	**Duration (Lecture)**
	1. Build reusable and interactive front-end components using frameworks like Vue.js, Svelte, Angular, or plain JavaScript, demonstrating state management and API integration. a. Develop UI components like forms, modals, or charts using Vue.js, Svelte, Angular, or plain JavaScript or similar b. Use state management tools like Vuex, Redux, or Context API for data handling c. Consume APIs (REST/GraphQL) to display real-time data and implement routing/authentication 2. Explore and implement API development using GraphQL or REST with Express or FastAPI, enabling efficient querying and data handling for complex relationships a. Set up a GraphQL API using Python's Graphene library or REST API using Flask/Django b. Create queries, mutations, or endpoints for nested data structures c. Test API functionality using tools like GraphiQL, Insomnia, or Postman 3. Design and implement authentication systems integrating front-end and back-end, with role-based access control (RBAC) a. Create login and register UIs with JWT or session-based authentication b. Secure API routes and add RBAC for specific user roles c. Implement advanced features like "Remember Me" and password resets 4. Integrate real-time communication in web applications using WebSocket-based tools for features like chat and live notifications a. Build a real-time chat application or live notifications system b. Use WebSocket events (e.g., user typing, message received) c. Establish communication between the front-end and back-end using WebSocket or tools like Firebase Realtime Database 5. Prepare the development environment for the MERN or alternative stack a. Install Node.js, npm, and MongoDB locally or set up Atlas for MongoDB b. Scaffold front-end projects using tools like Angular CLI, React's Create App, or Vue CLI	

c. Initialize a back-end server with Node.js and Express

6. Design efficient database schemas using NoSQL or SQL and apply CRUD operations
   a. Design schemas with tools like Mongoose or Sequelize ORM
   b. Add validations and constraints
   c. Perform CRUD operations on the database
7. Establish communication between the front-end and back-end using API calls
   a. Fetch and display data from APIs using Axios or Fetch API
   b. Handle errors and loading states effectively
   c. Use dynamic data binding to update components
8. Implement routing and navigation features in front-end applications with access controls and optimization techniques
   a. Set up routing with Angular Router or React Router
   b. Implement lazy loading for optimized performance
9. Deploy full-stack web applications on hosting platforms and manage configurations securely
   a. Host the front-end on platforms like Vercel, Netlify, or Firebase
   b. Deploy the back-end on Heroku, AWS, or Render
   c. Use environment variables to manage sensitive data
10. Testing the functionality of full-stack applications using testing frameworks for unit, integration, and end-to-end testing
    a. Write tests using Jest, Mocha, or Cypress for both front-end and back-end components
11. Explore server-side rendering for improved SEO and faster initial load times
    a. Set up SSR using frameworks like Next.js or Nuxt.js
    b. Render dynamic content on the server before serving it to the client
    c. Compare performance metrics between SSR and traditional CSR (Client-Side Rendering)
12. Exploring pyscript for FSD

Optional - Additional practical for assignment

1. Integrate AI/ML-powered features into web applications
   a. Use TensorFlow.js or ML5.js for client-side ML models
   b. Build features like text classification, sentiment analysis, or image recognition

	c. Fetch predictions from server-side ML models hosted on platforms like AWS SageMaker 2. Implement microservices architecture for a full-stack application using Docker to run services like user management, payment gateways, or analytics separately	
Total Lectures		**30**

<table>
<tr><td colspan="2">Program: B.Sc. – Information Technology (2026 - 27)</td><td colspan="2">Semester: V</td></tr>
<tr><td colspan="2">Course: Software Project Management</td><td colspan="2">Course Code: UISPM304</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td>Lecture (Hours per week)</td><td>Credit</td><td>Continuous Assessment (CA) (Marks - 20)</td><td>Semester End Examinations (SEE) (Marks- 30 in Question Paper)</td></tr>
</table>

2	2	20	30

Learning Objectives:

- To learn the basic concepts of Software Project Management
- To learn the concept of Resource allocation and activity planning
- To learn the importance of monitoring and control using Jira software

Course Outcomes:

After completion of the course, learners would be able to:

CO1: To understand the key concepts of Software Project Management

CO2: To identify the components of project planning and analyse the stepwise approach to project planning.

CO3: To analyse effort estimation, risk, and resource allocation, and their impact on project success.

CO4: To evaluate resource allocation strategies, identifying resource requirements and creating cost schedules.

CO5: To apply tools and techniques for prioritizing monitoring, implementing change control, and conducting project termination reviews.

CO6: To design strategies for monitoring progress, managing changes, and ensuring quality in software projects.

Outline of Syllabus: (per session plan)

Modules	Topics	Duration (Lecture)
Module 1	**Introduction to Software Project Management**	**10**
	Introduction, what is a Project, Activities Covered by Software Project Management, Plans, Methods and Methodologies, Project Planning: Introduction to Step Wise Project Planning Software Effort Estimation: Introduction, Problems with Over- and Under-Estimates, Th e Basis for Software Estimating, Software Eff ort Estimation Techniques	
Module 2	**Activity planning and Resource allocation**	**10**
	Introduction, Objectives of Activity Planning, Project Schedules, Projects and Activities, Sequencing and Scheduling Activities, Network Planning Models, Identifying the Critical Path, Risk management, PERT Technique, Monte Carlo Simulation Resource allocation: Identifying Resource Requirements, Scheduling Resources, Creating Critical Paths, Counting the Cost, Cost Schedules	
Module 3	**Monitoring and Control**	**10**

	Creating the Framework, Collecting the Data, Review, Visualizing Progress, Cost Monitoring, Earned Value Analysis, Change Control, Software Configuration Management (SCM) Project management with Jira: The JIRA architecture, The JIRA hierarchy, Project Administration, JIRA issue summary, Time tracking, Mapping business processes, understanding workflows, Managing workflows AI Application: Miro AI and LucidChart AI in UML Diagrams	
Total Lectures		**30**

Reference books:

1. Project management absolute beginner's guide, Gregory M Horine, 5th edition, Pearson Education, 2022
2. Jira 8 Essentials: Effective project tracking and issue management, Patrick Li, 6th Edition, Packt Publishing, 2022
3. Hughes, Bob, Mike Cotterell and Rajib Mall, Software Project Management (SIE), 6th Edition, New York:
4. McGraw Hill Education, 2017

Program: B.Sc. – Information Technology (2026 - 27)		**Semester: V**	
Course: Software Project Management LAB		**Course Code: UISPM304P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA)** **(Marks - 20)**	**Semester End Examinations (SEE)** **(Marks- 30 in Question Paper)**
2	1	20	30

Learning Objectives:

• To understand the importance of software testing in the software development lifecycle.
• To learn different software testing techniques such as black-box, white-box, and functional testing.
• To develop skills in preparing test cases, test scenarios, and test plans.
• To gain practical exposure to defect tracking and test documentation.
• To understand automation testing concepts using modern testing tools.how UML diagrams help in analyzing, designing, and documenting software systems.

<table>
<tr><td colspan="3"></td></tr>
<tr><td colspan="3">Course Outcomes:

• Students will be able to design and execute software test cases for validating system functionality.
• Students will be able to apply black-box and white-box testing techniques to identify software defects.
• Students will be able to prepare test plans, test scenarios, and maintain testing documentation.
• Students will be able to perform defect tracking and reporting using standard methods.
• Students will be able to understand and demonstrate basic automation testing using testing tools.</td></tr>
<tr><td colspan="3">Outline of Syllabus:</td></tr>
<tr><td></td><td>Practicals on Software Testing</td><td>Duration (Lecture)</td></tr>
<tr><td></td><td>Study of Software Testing Life Cycle (STLC) and preparation of a Test Plan.

Writing Test Cases and Test Scenarios for a given application (Login / Registration Module).

Black Box Testing using Equivalence Partitioning Technique.

Black Box Testing using Boundary Value Analysis.

White Box Testing – Control Flow Graph and Cyclomatic Complexity.

Unit Testing using JUnit or similar testing framework.

Integration Testing using Top-Down and Bottom-Up approaches.

System Testing and Functional Testing of a Web Application.

Defect Tracking and Bug Report Preparation using Bug Tracking Tools.</td><td></td></tr>
<tr><td colspan="2">Total Lectures</td><td>30</td></tr>
</table>

Reference Books

1. **Software Engineering: A Practitioner's Approach** – Roger S. Pressman and Bruce R. Maxim, **9th Edition**, McGraw-Hill Education, 2023.

2. **Software Engineering** – Ian Sommerville, **10th Edition**, Pearson Education, 2017.

3. **Fundamentals of Software Engineering** – Rajib Mall, **5th Edition**, PHI Learning, 2018.

4. **The Art of Software Testing** – Glenford J. Myers, Corey Sandler, and Tom Badgett, **3rd Edition**, Wiley, 2011.

5. **Effective Software Testing: A Developer's Guide** – Maurício Aniche, Manning Publications, 2022.

Programme : B.Sc. – Information Technology (2026 - 27)			**Semester V**
Course :Artificial Intelligence			**Code : UIARI305**
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA) (Marks - 40)**	**Semester End Examinations (SEE) (Marks- 60 in Question Paper)**
3	**3**	**40 Marks**	**60 Marks**

Learning Objectives

1. Understand the fundamentals of Artificial Intelligence (AI)
2. Learn about intelligent agents
3. Analyze problem-solving techniques in AI
4. Explore knowledge representation and reasoning.
5. Introduce neural networks and their applications.
6. Familiarize with AI applications.

Learning Outcomes:

1. The learner will gain knowledge of Artificial Intelligence and its foundations
2. The learner will identify basic principles of AI in problem solving
3. The learner will apply problem-solving methods and search strategies to AI problems
4. The learner will represent and infer knowledge using logical frameworks
5. Learner will be able to design and evaluate AI-based solutions.
6. Learner will be able to demonstrate the ability to apply AI techniques to real-world problems

Module	**Module Content**	**Duration**
I	Artificial Intelligence (AI), AI Perspectives: Acting and Thinking humanly, Acting and Thinking rationally History of AI, Applications of AI, The present state of AI, Ethics in AI. **Introduction of agents**, Structure of Intelligent Agent, Characteristics of Intelligent Agents. Types of Agents: Simple Reflex, Model Based, Goal Based, Utility Based Agents.	15

	Environment Types: Deterministic, Stochastic, Static, Dynamic, Observable, Semi-observable, Single Agent, Multi Agent **Solving Problems by Searching** Uninformed Search: Depth First Search, Breadth First Search, Informed Search: Heuristic Function, Greedy Best First Search, A* Search, Local Search: Hill Climbing Search Optimization: Genetic Algorithm, Mini-max Search	
II	**Introduction to Evolutionary Computation:** Elements of Genetic Algorithms, genetic programming concepts, Evolutionary programming, swarm intelligence Genetic Algorithms and Traditional Search Methods, Some Applications of Genetic Algorithms . **Knowledge and Reasoning** Definition and importance of Knowledge, Issues in Knowledge Representation, Knowledge Representation Systems, Properties of Knowledge Representation Systems	15
III	**Propositional Logic (PL):** Syntax, Semantics, Formal logic-connectives, truth tables, tautology, validity, well-formed-formula, Introduction to logic programming (PROLOG) **Predicate Logic**: FOPL, Syntax, Semantics, Quantification, Inference rules in FOPL, Forward Chaining, Backward Chaining and Resolution in FOPL. **Introduction to ANN** Units in neural networks, Network structures, Single layer feed-forward neural networks (perceptrons) Applications of ANN Pattern Classification, Associative memories, Optimization, Vector quantization, Control applications, Application in Speech and Image Processing	15
	Total Lectures	45

<u>Books and References:</u>

Sr. No .	Title	Author/s	Publisher	Edition	Year
1.	Artificial Intelligence- A Modern Approach	Stuart Russel, Peter Norvig	Pearson Education	4th	2020
2.	Artificial Intelligence	Saroj Kaushik	Cengage	1 st	2019

Useful Links
1 An Introduction to Artificial Intelligence - Course (nptel.ac.in)
2 NPTEL
3 https://www.classcentral.com/course/independent-elements-of-ai-12469
4 https://tinyurl.com/ai-for-everyone

Program: B.Sc. – Information Technology (2026 - 27)		Semester: V	
Course: Artificial Intelligence LAB		**Course Code: UIARI305P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA)** **(Marks - 20)**	**Semester End Examinations (SEE)** **(Marks- 30 in Question Paper)**
2	1	20	30

Learning Objectives:

🎬 To understand the basic concepts and techniques used in Artificial Intelligence problem solving.

🎬 To learn different search strategies including uninformed and informed search methods.

🎬 To understand game playing strategies using algorithms such as the MiniMax algorithm.

🎬 To explore optimization and evolutionary techniques such as Genetic Algorithms.

🎬 To gain practical knowledge of machine learning concepts by implementing Neural Networks and simple AI chatbots.

Course Outcomes:

🎬 Students will be able to implement uninformed and informed search algorithms to solve AI problems.

🎬 Students will be able to apply the MiniMax algorithm for game simulations such as Tic-Tac-Toe.

🎬 Students will be able to implement optimization techniques like Genetic Algorithms for problem solving.

🎬 Students will be able to develop simple Neural Network models using Python with and without built-in libraries.

🎬 Students will be able to design basic AI-based conversational agents using AIML packages.

Outline of Syllabus:		
	List of Practicals :	**Duration (Lecture)**
	1. Implementation of any uninformed search methods. 2. Simulation of TIC-TAC-TOE using MiniMax algorithm 3. Implementation of informed search methods. 4. Implementation of genetic algorithms methods. 5. Implementation of a simple Neural Network using python's inbuilt libraries. 6. Implementation of a simple Neural Network without using python inbuilt libraries. 7. Creating a simple bot using AIML package.	
Total Lectures		**30**

Program: B.Sc. – Information Technology (2026 - 27)	**Semester: V**
Course: Security in Computing	**Course Code: UISIC306**

Teaching Scheme		Evaluation Scheme	
Lecture **(Hours per week)**	**Credit**	**Continuous Assessment (CA)** **(Marks - 40)**	**Semester End Examinations (SEE)** **(Marks- 60 in Question Paper)**
3	3	40	60

Learning Objectives:

- To identify common programming flaws (buffer overflow, improper input validation, race conditions).
- To make the students aware about network security threats, attack vectors, and defensive mechanisms across wired and wireless environments.
- To help the students automate a complete process

Course Outcomes:

After completion of the course, learners would be able to:

CO1: Analyze fundamental concepts of computer security including threats, vulnerabilities, risks, and impact assessment.
CO2: Apply cryptographic principles and primitives to design secure communication mechanisms.
CO3: Assess database security requirements including reliability, integrity, recovery, concurrency control, and disclosure prevention mechanisms
CO4: Analyze network security threats, attack vectors, and defensive mechanisms across wired and wireless environments.
CO5: Evaluate correctness and completeness in OS security enforcement.

CO6: Analyze cloud computing architectures and evaluate associated security risks across service and deployment models.

CO7: Evaluate security risks in Generative AI systems

Outline of Syllabus: (per session plan)

Modules	Topics	Duration (Lecture)
Module I	**Introduction to Security, Program Security**	
	Introduction: What Is Computer Security?, Threats, Harm, Vulnerabilities.	**15**

	Program Security: Unintentional (Nonmalicious) Programming, Malicious Code—Malware, Countermeasures **Encryption**: Cryptology, Cryptanalysis Cryptographic Primitives One-Time Pads, Statistical Analysis ,What Makes a "Secure" Encryption Algorithm?, Symmetric Encryption Algorithms: DES, AES, RC2, RC4, RC5, and RC6 , Asymmetric Encryption with RSA: The RSA Algorithm, Strength of the RSA Algorithm, Message Digests: Hash Functions, One-Way Hash Functions, Message Digests, Digital Signatures: Elliptic Curve Cryptosystems, El Gamal and Digital Signature Algorithms, Quantum Cryptography: Quantum Physics, Photon Reception, Cryptography with Photons, Implementation	
Module II	Operating Security, Database Security and Network Security	**15**
	Operating Security: Security in Operating Systems Background: Operating System Structure Security Features of Ordinary Operating Systems A Bit of History Protected Objects Operating System Tools to Implement Security Functions, **Security in the Design of Operating Systems** : Simplicity of Design Layered Design Kernelized Design, Reference Monitor Correctness and Completeness Secure Design Principles Trusted Systems Trusted System Functions The Results of Trusted Systems Research, **Rootkit**: Phone Rootkit Rootkit Evades Detection Rootkit Operates Unchecked Sony XCP Rootkit TDSS Rootkits **Database Security**: Security Requirements of Databases, Reliability and Integrity: Recovery, Concurrency/Consistency, Database Disclosure: Sensitive Data, Types of Disclosures, Preventing Disclosure: Data Suppression and Modification: Security Versus Precision **Network** Security: Network Concepts Background: Network Transmission Media, Protocol Layers, Addressing and Routing Network Security Attacks, Threats to Network Communications Interception: Eavesdropping and Wiretapping Modification, Fabrication: Data Corruption, Interruption: Loss of Service, Port Scanning, Vulnerability	

	Summary, Wireless Network Security: WiFi Background, Vulnerabilities in Wireless Networks, Failed Countermeasure: WEP (Wired Equivalent Privacy), Stronger Protocol Suite: WPA (WiFi Protected Access), Denial of Service: Massive Estonian Web Failure, How Service Is Denied	
Module III	**Cloud Computing:** Cloud Computing Concepts: Service Models, Deployment Models; Moving to the Cloud: Risk Analysis, Cloud Provider Assessment, Switching Cloud Providers, Cloud as a Security Control, Cloud Security Tools and Techniques: Data Protection in the Cloud, Cloud Application Security, Logging and Incident Response, Cloud Identity Management: Security Assertion Markup Language, OAuth, OAuth for Authentication, Securing IaaS: Public IaaS Versus Private Network Security **Security Generative AI:** GenAI Data Security, GenAI Model Security **IOT Security:** Security and IoT: Overview of Cyberattacks in IoT, Distributed Denial of Service, Hardware Security, Hardware Security v/s Hardware Trust, Embedded System Hardware, Data Layers	**15**
Total Lectures		**45**

Books

- Charles P. Pfleeger Shari Lawrence Pfleeger Jonathan Margulies, Security in Computing FIFTH EDITION, 2015
- DATABASE PRINCIPLES AND TECHNOLOGIES – BASED ON HUAWEI GAUSSDB publication by Springer, 2023
- Ron Ben Natan, Implementing Database Security and Auditing A guide for DBAs, information security administrators and auditors, Elsevier Digital Press, 2005
- Ken Huang **, Generative AI Security: Theories and Practices by Springer publications, 2024**
- IoT SECURITY GUIDE, by Ministry of Electronics and Information Technology Government of India, DSCI (Nascom Initiative), August 2022
- Securing generative AI by IBM Institute for Business Value | Research Insights , IBM,AWS

Program: B.Sc. – Information Technology (2026 - 27)	**Semester: V**
Course: Security in Computing Practical	**Course Code: UISIC306P**
Teaching Scheme	**Evaluation Scheme**

Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks - 20)	Semester End Examinations (SEE) (Marks- 30)
2	1	20	30

Learning Objectives:

- To identify common programming flaws (buffer overflow, improper input validation, race conditions).
- To make the students aware about network security threats, attack vectors, and defensive mechanisms across wired and wireless environments.
- To help the students automate a complete process

Course Outcomes:

After completion of the course, learners would be able to:

CO1: Analyze fundamental concepts of computer security including threats, vulnerabilities, risks, and impact assessment.
CO2: Apply cryptographic principles and primitives to design secure communication mechanisms.
CO3: Assess database security requirements including reliability, integrity, recovery, concurrency control, and disclosure prevention mechanisms
CO4: Analyze network security threats, attack vectors, and defensive mechanisms across wired and wireless environments.
CO5: Evaluate correctness and completeness in OS security enforcement.

CO6: Analyze cloud computing architectures and evaluate associated security risks across service and deployment models.

CO7: Evaluate security risks in Generative AI systems

Outline of Syllabus: (per session plan)

Modules	Topics	Duration (Lecture)
	1. Malware Detection Using Hash Comparison 2. Trojan-like File Copy Simulation 3. Generate Hash for a Message 4. To compare DES and AES symmetric encryption algorithms. 5. To implement a simple access control enforcement program demonstrating the **Reference Monitor** principle. 6. To simulate a conceptual Denial of Service (DoS) scenario in a controlled lab environment and implement mitigation using Access Control Lists (ACLs). 7. Packet Tracer - Configure AAA Authentication on Cisco Routers	

	8. Configuring IPv6 ACLs 9. To implement Role-Based Access Control (RBAC) in a database system to restrict user privileges and enforce least privilege principle. 10. To simulate OAuth-style token-based authentication. 11. GenAI Data Poisoning Simulation 12. GenAI Data Poisoning Simulation 13. IoT-based DDoS Concept Simulation	
Total Lectures		**30**

Program: B.Sc. – Information Technology (2026 - 27)		**Semester: V**	
Course: Internet of Things		**Course Code: UIIOT307**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA) (Marks - 40)**	**Semester End Examinations (SEE) (Marks- 60 in Question Paper)**
3	3	20+20	60

Learning Objectives:

- Knowledge about the core concepts of IoT, its architecture, wearable devices
- Knowledge about usage of Raspberry Pi and ESP 32 in creation of IoT circuits
- Understanding the integration of IoT in various industries

Course Outcomes:

1. Remember the basic concepts of physical design, logical design, IoT levels and deployment models, Raspberry Pi, ESP32
2. Understand the IoT platform design methodology, Domain specific IoT's, IoT and Wearable Technology Architecture and Hardware
3. Application of Raspberry Pi and ESP 32 to interface with different sensors and actuators
4. Analyzing the integration of IoT in different industries
5. Creation of circuits using Raspberry Pi and ESP 32 using camera and display

Outline of Syllabus: (per session plan)

Modules	Topics	Duration (Lecture)
Module 1	**Introduction to Internet of Things**	**15**
	Introduction, physical design, logical design, IoT levels and deployment models, Domain specific IoT's – Home Automation, Environment, Energy, Retail, Logistics, Agriculture, Health and Lifestyle, IoT platform design methodology	
Module 2	Raspberry Pi and IoT in industry	**15**
	What Is the Raspberry Pi?, What Can You Do with a Raspberry Pi?, A Tour of the Raspberry Pi, Setting Up Your Raspberry Pi, Buying What You Need Connecting Everything Together, Booting Up Linux, The Desktop, The Internet, The Command Line, navigating with the Terminal, sudo, Applications, Internet Resources, Basic Python required for Raspberry pi programming Interfacing Hardware GPIO Pin Connections, Pin Functions, Serial Interface Pins, Power Pins Hat Pins, Breadboarding with Jumper Wires, Digital Outputs Analog Outputs, Digital Inputs, Analog Inputs, Hardware, The Software IoT in industry Manufacturing, Oil and Gas, Utilities, Smart and Connected Cities, Transportation, Public Safety	

Module 3	ESP 32	15
	ESP 32 Understanding the capabilities of ESP32 for IoT, Deep dive into the Arduino IDE 2.0 to program ESP32, Connecting Sensors and Actuators with ESP32, Getting hands-on with ESP32 GPIO pins and an overview of them, Mastering UART communication, I2C communication with ESP32, Understanding SPI communication, Interfacing Cameras and Displays with ESP32, Using the ESP32 camera module, Interfacing displays with ESP32, Comparison of displays. Wearable devices IoT and Wearable Technology Enabled Applications, Architectures- IoT and Wearable Technology Architectures, The Motivations Behind New Architectures, Edge Computing, Cloud, Fog, and Mist, IoT Architectures, Common IoT Architectures, 4 Layers, Wearable Device Architecture Hardware- Hardware Components Inside IoT and Wearable Devices, Sensors, Wireless Sensors, Multisensor Modules AI tools – Blynk, OpenCV	
Total Lectures		**45**

Reference books:

Hands-on ESP32 with Arduino IDE, Asim Zulfiqar , 2023 Packt Publishing

Fundamentals of IoT and Wearable Technology Design, Haider Raad, Copyright © 2021

IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017

Programming the Raspberry Pi, Second Edition: Getting Started with Python, Simon Monk, 2016

“Internet of Things (A Hands-on Approach)”, 1st Edition, VPT,Vijay Madisetti and Arshdeep

Bahga, 2014

Program: B.Sc. – Information Technology (2026 - 27)		**Semester: V**	
Course: Internet of Things LAB		**Course Code: UIIOT307P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA) (Marks -)**	**Machine Test**
2	1	20	30

Learning Objectives:

• To understand the fundamental concepts of the Internet of Things (IoT) and embedded systems.
• To gain hands-on experience with IoT hardware platforms such as Raspberry Pi and ESP32.
• To learn interfacing of sensors, actuators, and peripheral devices with IoT boards.
• To develop the ability to write programs in Python and embedded C for hardware interaction.
• To design and implement simple IoT-based automation and monitoring systems.

Course Outcomes:

• Students will be able to configure and use Raspberry Pi and ESP32 for IoT applications.
• Students will be able to interface sensors, displays, and communication modules with IoT devices.
• Students will be able to develop programs for controlling hardware components using GPIO.
• Students will be able to design basic IoT solutions for real-world applications such as home automation and monitoring systems.
• Students will be able to integrate hardware and software components to build functional IoT prototypes.

Topics

Raspberry Pi Practical

1. Raspberry Pi Hardware Preparation and Installation
2. GPIO: Light the LED with Python
3. Displaying different LED Patterns with Raspberry Pi
4. Displaying Time over 4 Digit 7 Segment Display using Raspberry Pi
5. Fingerprint Sensor interfacing with Raspberry Pi
6. GPS Module interfacing with Raspberry Pi
7. RFID Module interfacing with Raspberry Pi
8. Capturing Images with Raspberry Pi and Pi Camera
9. Visitor Monitoring with Raspberry Pi and Pi Camera
10. IOT Based Home Automation using raspberry Pi
11. Installing Windows 10 IOT Core on Raspberry Pi
12. Building Google Assistant with Raspberry Pi
13. Setting up Wireless Access Point using Raspberry Pi

ESP 32 Practical

1. Blinking A Led
2. Controlling Led Using Switch

3. Seven Segment Displays 4. Liquid Crystal Display 5. Temperature Sensor 6. Stepper Motor

Reference books:

Hands-on ESP32 with Arduino IDE, Asim Zulfiqar , 2023 Packt Publishing

Fundamentals of IoT and Wearable Technology Design, Haider Raad, Copyright © 2021

IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017

Programming the Raspberry Pi, Second Edition: Getting Started with Python, Simon Monk, 2016

“Internet of Things (A Hands-on Approach)”, 1st Edition, VPT,Vijay Madisetti and Arshdeep

Bahga, 2014

Program: B.Sc. – Information Technology (2026 - 27)	**Semester: V**
Course: Android Programming LAB	**Course Code: UIANP308P**
Teaching Scheme	**Evaluation Scheme**

Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks -20)	Semester End Examinations (SEE) (Marks- 30 in Question Paper)
2	1	20	30

Learning Objectives:

🎬 To understand the fundamentals of the Android operating system and Android Studio development environment.

🎬 To learn how to design and develop basic Android applications using activities, fragments, and layouts.

🎬 To gain knowledge of Android UI components and resources such as menus, dialogs, and themes.

🎬 To understand event handling, intents, services, and broadcast receivers in Android applications.

🎬 To develop mobile applications that interact with databases and multimedia components such as SQLite/Firebase, audio, and video.

Course Outcomes:

🎬 Students will be able to create and run basic Android applications using Android Studio.

🎬 Students will be able to design user interfaces using various Android layouts and UI components.

🎬 Students will be able to implement activity lifecycle, fragments, and navigation using intents.

🎬 Students will be able to handle events, services, notifications, and broadcast receivers in Android apps.

🎬 Students will be able to develop Android applications that store data using SQLite/Firebase and support multimedia features.

Outline of Syllabus:

Topics

1. Introduction to Android, Introduction to Android Studio IDE, Application

2. Fundamentals: Creating a Project, Simple “Hello World” program.

3. Programming Resources Android Resources: (Color, Theme, String, Drawable, Dimension, Image),
    a.
4. Programming Activities and fragments Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.

5. Programs related to different Layouts Coordinate, Linear, Relative, Table, Absolute, Frame, List View, Grid View.

6. Programming UI elements AppBar, Fragments, UI Components

7. Programming menus, dialog, dialog fragments

8. Programs on Intents, Events, Listeners and Adapters

9. Programs on Services, notification and broadcast receivers

10. Database Programming with SQLite/Firebase

11. Programming with Media (Audio, Video)

Program: B.Sc. – Information Technology (2026 - 27)		**Semester: V**	
Course: Enterprise Java LAB		**Course Code: UIENJ302P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA)**	**Semester End Examinations (SEE)**

2	1	20	30

Learning Objectives:

Recall the fundamental concepts of Enterprise Java technologies including Servlets, JSP, JDBC, JSF, and Enterprise Beans.
Explain the working principles of web-based enterprise applications using Java technologies such as sessions, cookies, filters, and request dispatching.
Apply Java EE programming techniques to develop web components such as Servlets and JSP pages for handling client requests and responses.
Analyze the use of session tracking mechanisms and database connectivity in enterprise web applications.
Evaluate different Java EE technologies such as JSF, EJB, and JPA for building scalable enterprise applications.
Design and develop enterprise-level applications by integrating technologies like Servlets, JSP, WebSocket, JSON processing, and AI chatbot interfaces.

Course Outcomes:

After completion of the course, learners would be able to:

CO1: Describe the basic architecture and components of Enterprise Java applications.

CO2: Develop web applications using Servlets, JSP, and JDBC for handling user requests and database operations.

CO3: Implement session management, cookies, filters, and request dispatching techniques in Java web applications.

CO4: Apply advanced enterprise technologies such as JSF, EJB, and JPA to create dynamic and database-driven applications.

CO5: Analyze and integrate modern communication technologies such as WebSocket and JSON in enterprise applications.

CO6: Design and build a complete enterprise web application incorporating database connectivity and AI-based chatbot integration.

Outline of Syllabus:

List of Practicals

Servlet and JDBC Based Practicals

1. **Welcome Servlet**
   Implement a basic **WelcomeServlet** that prints a greeting message to the user.
2. **Servlet with HTML Form**
   Create a servlet that accepts user input from an HTML form and displays the submitted data.

3. **Servlet Using GET and POST Methods**
   Develop a servlet that demonstrates the use of **doGet() and doPost()** methods.
4. **Database Connectivity using JDBC**
   Write a Java servlet program that **connects to a database using JDBC** and retrieves records from a table.
5. **Insert Data into Database using Servlet**
   Create a servlet that accepts form data and **stores the data into a database table**.

Cookies and Session Management, Request Dispatcher

6. **Cookie Creation and Retrieval**
   Develop a servlet that **creates a cookie**, sets an expiry time, and retrieves the cookie on the next request.
7. **Session Management using HttpSession**
   Create a servlet that **creates an HTTP session and stores user information**.
8. **Session Tracking Application**
   Develop a servlet that demonstrates **session tracking using URL rewriting**.
9. **Session Login Application**
   Build a simple **login application using sessions** where only authenticated users can access a protected page.

10. **Request Dispatcher Example**
    Write a servlet program demonstrating **forward() and include() methods of RequestDispatcher**.

JSP Practicals

11. **Basic JSP Page**
    Create a JSP page displaying **"Hello World"** using JSP elements.
12. **JSP Directives Example**
    Develop a JSP page demonstrating the **page directive, include directive, and taglib directive**.
13. **JSP Scripting Elements**
    Create a JSP page using **declaration, scriptlet, and expression tags**.
14. **JSP Implicit Objects**
    Write a JSP program demonstrating the use of **request, response, session, application, and out objects**.
15. **JSP Action Tags**
    Develop a JSP application using **jsp:include, jsp:forward, and jsp:useBean**.
16. **Session Bean Application (EJB)**
    Develop a **Stateless Session Bean** that performs arithmetic operations and invoke it from a client.

Reference books:

1. Java: The Complete Reference by Herbert Schildt (Author), Dr. Coward, Danny (Author) ,McGraw-Hill Osborne Media; 13th edition (11 January 2024)

Web References :

https://docs.oracle.com/javaee/7/tutorial/index.html
https://docs.oracle.com/javaee/7/tutorial
https://jakarta.ee/learn/
https://www.w3schools.com/java/default.asp
https://www.w3schools.com/
https://www.tutorialspoint.com/swing/index.htm
https://www.tutorialspoint.com/
https://hibernate.org/
https://struts.apache.org/
https://www.geeksforgeeks.org/java/introduction-and-working-of-struts-web-framework/
https://www.geeksforgeeks.org

Programme : B. Sc. Information Technology (2026 - 27)			Semester V
Course : Field Project			Code : UIFPR310
Teaching Scheme		Evaluation Scheme	
Lecture	Credits		
		Continuous Assessment (CA) (Marks - 20)	Semester End Examinations (SEE) (Marks- 30)
02	2	**20**	**30**

SEM III/IV/V – FIELD PROJECT/COMMUNITY ENGAGEMENT PROJECT

COURSE CREDITS: 02 (All Programs)

Introduction

Inclusion of project work in the course curriculum of the B.Com. Programme is one of the ambitious aspects in the programme structure under NEP. The main objective of inclusion of project work is to inculcate the element of research work challenging the potential of learner as regards to his/her eagerness to enquire and ability to interpret particular aspect of the study in his/her own words. It is expected that the guiding teacher should undertake the counselling sessions and make the awareness among the learners about the methodology of formulation, preparation and evaluation pattern of the project work.

1. Credit Allocation:

Field Project   1 credit = 30 hours (2 Credits = 60 Hours)

Community Engagement Project   1 Credit = 45 Hours (2 credits = 90 hours)

2. Evaluation Pattern (Total 50 Marks):

- Report (30 Marks):
  - Must be submitted as a bound report with detailed documentation.
- Viva Voce (20 Marks):
  - To be conducted by the assigned faculty member in charge.

Page 2

Guidelines for preparation of Project Work under Field Project and Community Engagement Project

1. The field Project/Community Engagement Project may be undertaken in any areas of Major subject.
2. The Project has to be done in Teams of two learners each, which would be formed by the Teacher-Guide/s allotted by the respective Head of Dept./Co-ordinator for the respective programs.
3. Topics suggested by the Teams should be specific, clear and within definite scope of the concerned subject and have to be sanctioned by their respective Teacher-guides.
4. The project report shall be prepared as per the broad guidelines given below:

• Font type: Times New Roman
• Font size: 12 for content, 14 for Title
• Line Space: 1.5 for content and 1 for table work
• Paper Size: A4
• Margin: Left – 1.5, Up/Down/Right – 1
• The Project Report shall be bounded.
• The project report should be of 40 to 50 sides

Project Evaluation criteria (rubrics)
The Project Report shall be evaluated as follows:

1. Evaluation of Project Report (Bound Copy) – 30 Marks

□ Conceptual understanding, appropriate details of tasks and projects assigned – 10 Marks
□ Rationale of the Project and Practical insights derived from the project – 10 Marks
□ Overall structure of the Report – 10 Marks

2. Conduct of Presentation and Viva-voce – 20 Marks

□ Presentation by the Team – 10 Marks
□ In the course of Viva-Voce, questions may be asked based on conceptual understanding, conclusion and recommendation given by the Team – 10 Marks

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Format of Project Report

Shri Vile Parle Kelavani Mandal's

Usha Pravin Gandhi College of Arts, Science and Commerce

(Autonomous)

Title of the Project

A Project Submitted to

Department of Information Technology/Accountancy/Commerce/ under Field Project/Community Engagement Project in the programme Bachelor of Commerce/Science

By

Name of learners

Div

Roll nos.

SAP nos.

Under the Guidance of

Name of Teacher-Guide

Date, Month and Year

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Index

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Project should be divided as follows

- Part 1: Introduction Chapter 1:
- Part 2: Main Content

  This should be suitably divided in various chapters.
- Part 3: Conclusions and Suggestions

  In this chapter of project work, findings of work will be covered and suggestion will be enlisted.

Note: If required more chapters of data analysis can be added.

- Bibliography
- Plagiarism Report
- Appendix

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SHRI VILE PARLE KELAVANI MANDAL's

USHA PRAVIN GANDHI COLLEGE OF ARTS, SCIENCE AND COMMERCE

Autonomous – Affiliated to University of Mumbai

Bhakti Vedanta Swami Marg, Juhu Scheme, Vile Parle (West), Mumbai 400 056.

NAAC Re-accredited "A+" Grade with CGPA 3.27

CERTIFICATE

This is to certify that Ms/Mr ________________________ and Ms/Mr ________________________ have worked and duly completed their Field Project/community Engagement Project Work in B.Com Semester III/IV under the Department of Accountancy/Commerce in the subject of "________________________" (specialisation) and their project is entitled, "________________________" (Title of the Project) under my supervision (Name of Teacher-Guide). I further certify that the entire work has been done by the learners under my guidance and that no part of it has been submitted previously for any Degree or Diploma of any University.

It is their own work and facts reported by their personal findings and investigations.

Name and Signature of Guiding Teacher

[Seal of the College]

Date of submission:

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Declaration by learner

We the undersigned Ms/Mr ________________________ and ________________________, name of the learners hereby declare that the work embodied in this project work titled "________________________", Title of the Project, forms our own contribution to the project work carried out under the guidance of ________________________. It has not been previously submitted to any other University for any other Degree/Diploma to this or any other University.

Wherever reference has been made to previous works of others, it has been clearly indicated as such and included in the bibliography.

We hereby further declare that all information of this document has been obtained and presented in accordance with academic rules and ethical conduct.

Names and Signature of the learners

Certified by

Name and signature of the Guiding Teacher

Acknowledgement

To list who all have helped me is difficult because they are so numerous and the depth is so enormous.

We would like to acknowledge the following as being idealistic channels and fresh dimensions in the completion of this project.

We take this opportunity to thank SVKM's Usha Pravin Gandhi College of Arts, Science and Commerce for giving us chance to do this project.

We would like to thank Principal, for providing the necessary facilities required for completion of this project.

We take this opportunity to thank the Head of the Department, for her moral support and guidance.

We would also like to express our sincere gratitude towards our project guide whose guidance and care made the project successful.

We would like to thank College Library, for having provided various reference books and magazines related to our project.

Lastly, we would like to thank each and every person who directly or indirectly helped us in the completion of the project especially our Parents and Peers who supported us throughout our project.

Signature

HOD

Signature

Approved by Vice-Principal

Signature

Approved by Principal

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LOG BOOK

Field Project 60 hours / CEP 90 hours Log Report

Semester V Academic Year 2025–26

<table>
<tr><td>Project Title</td><td colspan="4"></td></tr>
<tr><td rowspan="3">Name of the students</td><td>Sr. No.</td><td>Full Name of Student</td><td>Class & Div.</td><td>Roll No.</td></tr>
<tr><td>1.</td><td></td><td></td><td></td></tr>
<tr><td>2.</td><td></td><td></td><td></td></tr>
</table>

Date	No. of Hour(s) spent	Name of the activity
Total	30 Hours	

Name & Signature of Students	Signature & name of the Project Guide

SEMESTER VI

Program: B.Sc. – Information Technology (2026 - 27)	Semester: VI
Course: Introduction to DevOps	**Course Code:** UIITD351
Teaching Scheme	**Evaluation Scheme**

Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks - 20)	Semester End Examinations (SEE) (Marks- 30 in Question Paper)
2	2	10+10	30

Learning Objectives:

- To learn the basic concepts of DevOps
- To learn the concept of version control, Containerization
- To learn the importance of Software Configuration Management

Course Outcomes:

After completion of the course, learners would be able to:

CO1: To understand the principles of DevOps culture, CI/CD pipelines, and containerized microservices.

CO2: To implement workflows using tools like Git, Terraform, Ansible, Docker, and Kubernetes to streamline software development and deployment.

CO3: To evaluate how infrastructure as code, CI/CD pipelines, and containerization enhance software delivery and operational efficiency.

CO4: To analyse different tools and technologies for their effectiveness in automating infrastructure provisioning and managing containerized applications.

CO5: To design and implement CI/CD pipelines to build, test, and deploy applications seamlessly across various environments.

CO6: To demonstrate the ability to troubleshoot, optimize, and manage DevOps workflows and containerized applications effectively.

Outline of Syllabus: (per session plan)

Modules	Topics	Duration (Lecture)
Module 1	**DevOps and Infrastructure as Code**	**10**
	The DevOps Culture and Infrastructure as Code Practices Getting started with DevOps, Implementing CI/CD and continuous deployment, Provisioning Cloud Infrastructure with Terraform, Using Ansible for Configuring IaaS Infrastructure, Optimizing Infrastructure Deployment with Packer, Authoring the Development Environment with Vagrant	
Module 2	**DevOps CI/CD Pipeline**	**10**
	Managing Your Source Code with Git: technical requirements, Overviewing Git and its principal command lines, Understanding the Git process and Gitflow pattern Continuous Integration and Continuous Deployment: CI/CD principles, Using a package manager in the CI/CD process, Using Jenkins for CI/CD implementation, Using Azure Pipelines for CI/CD	

	Deploying Infrastructure as Code with CI/CD Pipelines: Running Packer in Azure Pipelines, Running Terraform and Ansible in Azure Pipelines	
Module 3	**Containerized Microservices with Docker and Kubernetes**	**10**
	Containerizing Your Application with Docker: Installing Docker, Creating a Dockerfile, Pushing a Docker image to a private registry (ACR), Deploying a container to ACI with a CI/CD pipeline, Using Docker for running command-line tools, Deploying Docker Compose containers in ACI Managing Containers Effectively with Kubernetes: Installing Kubernetes, Using Helm as a package manager, Using AKS, Creating a CI/CD pipeline for Kubernetes with Azure Pipelines, Monitoring applications and metrics in Kubernetes **Application level AI : Automated Testing and CI/CD Optimization** ML-Based Test Prioritization: AI helps prioritize test cases based on historical failure patterns. AI-Driven CI/CD Pipelines: Tools like CircleCI and GitHub Actions use AI to optimize build times.	
Total Lectures		**30**

Reference books:

1. Learning DevOps: A comprehensive guide to accelerating DevOps culture adoption with Terraform, Azure DevOps, Kubernetes, and Jenkins, Mikael Krief, Second Edition, Packt Publishing, 2022
2. DevOps Tools from Practitioner's ViewPoint Deepak Gaikwad, Viral Thakkar Wiley -- 2019
3. Mitesh Soni, "DevOps Bootcamp", Packt Publishing Ltd, 2017.
4. Joakim Verona. Practical Devops, Second Edition. Ingram short title; 2nd edition (2018). ISBN10: 1788392574
5. The DevOps Adoption Playbook: A Guide to Adopting DevOps in a MultiSpeed IT Enterprise Sharma S Wiley First 2017

Program: B.Sc. – Information Technology (2026 - 27)		**Semester: VI**	
Course: Introduction to DevOps LAB		**Course Code: UIITD351P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA)** **(Marks - 20)**	**Semester End Examinations (SEE)** **(Marks- 30 in Question Paper)**
2	1	20	30
Learning Objectives: 🎬 To understand the fundamental concepts of DevOps practices and continuous integration/continuous deployment (CI/CD).			

🎬 To learn how to install, configure, and use DevOps tools such as Jenkins, Docker, and Kubernetes.

🎬 To gain practical experience in automating build, test, and deployment processes using CI/CD pipelines.

🎬 To understand containerization and orchestration concepts using Docker and Kubernetes.

🎬 To explore infrastructure automation and configuration management using tools such as Puppet and cloud platforms.

Course Outcomes:

🎬 Students will be able to set up and manage CI/CD pipelines using Jenkins and related build tools such as Maven, Gradle, or Ant.

🎬 Students will be able to develop and deploy containerized applications using Docker.

🎬 Students will be able to orchestrate containerized applications using Kubernetes.

🎬 Students will be able to implement configuration management and automation using Puppet.

🎬 Students will be able to deploy and monitor applications on cloud platforms using tools such as Azure DevOps and Google Cloud services

Outline of Syllabus:

	List of Practicals :	**Duration (Lecture)**
	1. Jenkins installation and setup, explore the environment and demonstrate continuous integration and development using Jenkins. 2. Explore Docker commands for content management. Develop a simple containerized application using Docker. 3. Integrate Kubernetes and Docker. Automate the process of running containerized application using Kubernetes. 4. Set up Azure DevOps. Import Code and Create the Azure DevOps Build Pipeline. Create the Azure DevOps Release Pipeline 5. To understand Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to setup a build Job. 6. To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server. 7. To understand Jenkins Master-Slave Architecture and scale your Jenkins standalone implementation by implementing slave nodes. 8. To Setup and Run Selenium Tests in Jenkins Using Maven.	

	9. To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers. 10. To learn Dockerfile instructions, build an image for a sample web application using Dockerfile. 11. To install and Configure Pull based Software Configuration Management and provisioning tools using Puppet. 12. To learn Software Configuration Management and provisioning using Puppet Blocks (Manifest, Modules, Classes, Function) 13. To provision a LAMP/MEAN Stack using Puppet Manifest. 14. Deploy a CI/CD pipeline using Google Cloud Build, integrate Kubernetes with GKE, automate infrastructure provisioning with Terraform, and monitor applications using Cloud Operations Suite.	
Total Lectures		**30**

Program: B.Sc. – Information Technology (2026 - 27)		**Semester: VI**	
Course: Data Science		**Course Code: UIDAS352**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA) (Marks - 40)**	**Semester End Examinations (SEE) (Marks- 60 in Question Paper)**
3	3	40	60
Learning Objectives 1. To develop the ability to distinguish and categorize different data types. 2. To cultivate the ability to analyze and interpret data visualizations and EDA. 3. To apply data transformation techniques for model preparation. 4. To evaluate and select appropriate machine learning algorithms. 5. To design and implement predictive models for real-world problems. 6. To integrate insights from unsupervised and supervised learning to solve business problems.			

Learning Outcomes:

1. Ability to distinguish and categorize various data types.
2. Ability to analyze and interpret data visualizations and perform EDA.
3. Ability to apply data transformation techniques for model preparation.
4. Ability to evaluate and select appropriate machine learning algorithms.
5. Ability to design and implement predictive models for complex problems.
6. Ability to integrate insights from unsupervised and supervised learning to solve real-world problems.

Outline of Syllabus:

Module	Module Content	Duration
I	**Introduction to Data Science:** What is Data? Different kinds of data, Quantitative vs. Qualitative data, Types of Quantitative data: Continuous data, Discrete data, Types of Qualitative data: Categorical data, Binary data, Ordinary data, Plotting data using Bar graph, Pie chart, Histogram, Scatter plot, Time-series graph, Exponential graph, Logarithmic graph, Frequency distribution graph **Exploratory Data Analysis (EDA)** Need of exploratory data analysis, cleaning and preparing data, Feature engineering, Missing values **Data transformations**: Dimension reduction, Feature extraction, Smoothing and aggregating. Principal Components Analysis (PCA) for Dimensionality Reduction	15
II	**Statistical Modelling and Machine Learning:** Regression (Linear / Logistic) Regularization, Ridge and LASSO regressions **Time series Concepts** - Univariate and multivariate. Stationary processes, ARMA models, spectral analysis, model and forecasting using ARMA models. Nonstationary and Seasonal Time Series Models, ARCH GARCH models, BASS Diffusion Model, Holt-Winters exponential smoothing. Modern Forecasting Models	15
III	**Supervised Learning**: Classification Problems: Definition, types, and real-world examples. Performance Evaluation Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC Curve Classification trees, Support Vector Machines, k-NN. Theory and solved sums **Applications**: Text, Image Classification **Unsupervised Learning**: Difference between supervised and unsupervised learning Types of problems addressed by unsupervised learning k-means clustering, Hierarchical clustering. Applications and use cases of clustering Customer Segmentation (e.g., e-commerce platform), Market Basket	15

	Analysis, Anomaly Detection (e.g., fraud detection) Document Clustering (e.g., news articles categorization) **H2O.ai –** AI-powered AutoML for machine learning and gradient boosting	

Books and References:

1) Doing Data Science, Rachel Schutt and Cathy O'Neil, O'Reilly,2013
2) Mastering Machine Learning with R, Cory Lesmeister, PACKT Publication,2015 **Additional Reference(s):**
1) Hands-On Programming with R, Garrett Grolemund,1st Edition, 2014
2) An Introduction to Statistical Learning, James, G., Witten, D., Hastie, T.,Tibshirani, R.,Springer,2015

Program: B.Sc. – Information Technology (2026 - 27)		**Semester: VI**	
Course: Data Science LAB		**Course Code: UIDAS352P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA)** **(Marks - 20)**	**Semester End Examinations (SEE)** **(Marks- 30 in Question Paper)**
2	1	20	30
Learning Objectives: 🎬 To understand the fundamental concepts of data science and data analysis techniques. 🎬 To learn methods for data collection, data preprocessing, and data management using modern database systems. 🎬 To gain practical knowledge of statistical analysis and hypothesis testing for data-driven decision making. 🎬 To explore machine learning algorithms such as regression, clustering, and classification techniques.			

🎬 To develop skills in implementing data science models using R or Python programming.

Course Outcomes:

🎬 Students will be able to collect, curate, and manage datasets using NoSQL databases.

🎬 Students will be able to apply dimensionality reduction and clustering techniques such as Principal Component Analysis and K-means.

🎬 Students will be able to build predictive models using regression techniques such as Simple Linear Regression and Logistic Regression.

🎬 Students will be able to apply statistical methods including hypothesis testing to analyze data.

🎬 Students will be able to implement machine learning algorithms such as KNN and Decision Trees using R or Python.

Outline of Syllabus:

	List of Practical (To be conducted using R/Python)	**Duration (Lecture)**
	1. Implementation of Data collection, Data curation and management (with any NoSQL DBMS) 2. Implementation of Principal Component Analysis 3. Implementation of k-means Clustering 4. Implementation of Simple Linear Regression 5. Implementation of Logistics Regression 6. Implementation of Hypothesis testing 7. Implementation of KNN 8. Implementation of Decision Tree	
	Total Lectures	**30**

Program: B.Sc. – Information Technology (2026 - 27)		Semester: VI	
Course: Cloud Computing		**Course Code: UICLC353**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 20)**	**End Semester Examinations (ESE) (Marks-30 in Question Paper)**
2	2	10+10	30

Learning Objectives:

- Define cloud computing and its key components.
- Understand the principles of parallel and distributed computing, and describe how these computing models contribute to the functionality of cloud systems.
- Demonstrate the use of virtualization techniques in cloud environments and evaluate their advantages and disadvantages within cloud architectures.
- Compare and contrast the features and functionalities of major cloud platforms like Amazon Web Services (AWS), Google App Engine, and Microsoft Azure, and identify their suitable applications.
- Develop and implement a cloud-based application for scientific and business purposes, integrating cloud computing and virtualization principles to meet specific requirements.
- Evaluate the economic, technical, and social implications of adopting cloud computing in various sectors, and analyze the open challenges such as scalability, security, and performance

Course Outcomes:

After completion of the course, learners would be able to:

1. Identify and describe the core concepts of cloud computing, including its vision, reference model, characteristics, and benefits.
2. Differentiate between parallel and distributed computing, explaining the elements that define each model and their relevance to cloud computing.
3. Apply virtualization techniques in the context of cloud environments, and assess their role in optimizing cloud infrastructure and services.
4. Analyze cloud computing architectures and compare the various cloud models (public, private, hybrid) and platforms (AWS, Google App Engine, Microsoft Azure).
5. Design cloud-based solutions for various scientific and business applications, integrating relevant technologies such as cloud computing and virtualization.
6. Evaluate the challenges, economic implications, and open issues in adopting cloud computing, including its impact on industries like healthcare, biology, geoscience, and business applications.

Outline of Syllabus: (per session plan)

Modules	Topics	Duration (Lecture)
Module 1	**Cloud Computing at a Glance:** The vision of Cloud computing , Defining a cloud,A closer look, Cloud computing reference model, Characteristics and Benefits. Principles of Parallel and Distributed Computing : Eras of Computing, Parallel v/s distributed computing, Elements of Parallel Computing, Elements of distributed computing.	**07**
Module 2	**Virtualization :** Introduction,Characteristics of Virtualized Environments,Taxonomy of Virtualization techniques, Virtualization and Cloud Computing ,Pros and cons of Virtualization **Cloud Computing Architecture:** Introduction,Cloud Reference Model,Types of Cloud,Economics of the cloud,open challenges.	**15**
Module 3	Introduction to Cloud Security. Cloud Platforms in Industry : Amazon web services,Google App Engine,Microsoft Azure Cloud Applications: Scientific Applications : i. Healthcare : ECG Analysis in the Cloud. ii. Biology : Protein Structure Prediction iii. Biology : Gene Expression Data Analysis for Cancer Diagnosis iv. Geoscience : Satellite Image Processing Business and Computer Applications using AI i. CRM & ERP ii. Productivity iii. Social Networking with AI for content generation iv. Media Applications v. Multiplayer Online Gaming.	**08**

Total Lectures	30

Reference books:

1. Foundations and Applications Programming Rajkumar Buyya, Christian , S. Thamarai Selvi, Mc Graw Hill ,Fifteenth Reprint 2019.
2. Mastering Cloud Computing | 2nd Edition Paperback – 1 June 2024, by Rajkumar Buyya (Author), Christian Vecciola (Author), Thamarai Selvi (Author), Shivananda Poojara (Author), Satish N. Srirama (Author) McGrawHill
3. Distributed and cloud computing from parallel processing to internet of things Paperback – 26 July 2017 by KAI HWANG / GEOFFREY C FOX / JACK J DONGARRA (Author)
4. **https://aws.amazon.com/training/**

<table>
<tr><td colspan="2">Program: B.Sc. – Information Technology (2026 - 27)</td><td colspan="2">Semester: VI</td></tr>
<tr><td colspan="2">Course: Cloud Computing LAB</td><td colspan="2">Course Code: UICLC353P</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td>Lecture
(Hours per week)</td><td>Credit</td><td>Continuous Assessment (CA)

(Marks - 20)</td><td>Semester End Examinations (SEE)

(Marks- 30 in Question Paper)</td></tr>
<tr><td>2</td><td>1</td><td>20</td><td>30</td></tr>
<tr><td colspan="4">Learning Objectives:

🎬 To understand the fundamentals of cloud computing platforms and web application deployment.

🎬 To learn how to develop web applications using MVC architecture with technologies such as Servlets and Spring Framework.

🎬 To explore different cloud service platforms such as Google App Engine, Microsoft Azure, and IBM Bluemix.

🎬 To gain knowledge of cloud storage and APIs for secure data management and integration.

🎬 To understand containerization and automation tools such as Docker, Chef, and Puppet used in cloud environments.</td></tr>
<tr><td colspan="4">Course Outcomes:

🎬 Students will be able to develop and deploy web applications using MVC frameworks and Java-based technologies.</td></tr>
</table>

🎬 Students will be able to set up and work with cloud platforms such as Google App Engine, Azure, and IBM Bluemix.

🎬 Students will be able to integrate cloud storage services using APIs such as Dropbox API.

🎬 Students will be able to implement containerization using Docker and manage multiple application images.

🎬 Students will be able to configure and deploy applications using automation tools such as Chef or Puppet in a cloud environment.

Outline of Syllabus:

	List of Practicals :	**Duration (Lecture)**
	1. Using the software like JDK 1.8, Eclipse IDE, Apache tomcat server 7.0 Servlets, Spring framework design and develop Web applications using MVC Framework. 2. Installing and configuring the required platform for Google App Engine 3. Studying the features of the GAE PaaS model 4. Developing an ASP.NET based web application on the Azure platform 5. Creating an application in Dropbox to store data securely. Develop a source code using Dropbox API for updating and retrieving files. 6. Installing Cloud Foundry in localhost and exploring CF commands. 7. Cloud application development using IBM Bluemix Cloud 8. Installing and Configuring Dockers in localhost and running multiple images on a Docker Platform. 9. Configuring and deploying VMs/Dockers using Chef/Puppet Automation tool.	
	Total Lectures	**30**

Program : B. Sc. IT (Information Technology) (2026-2027)			**Semester VI**
Course :Advanced Artificial Intelligence			**Code : UIAAI354**
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA) (Marks - 40)**	**Semester End Examinations (SEE) (Marks- 60 in Question Paper)**
3	**3**	**40 Marks**	**60 Marks**

Learning Objectives

By the end of this course, students will be able to:

1. Understand the fundamentals of soft computing and differentiate between soft computing and hard computing techniques.
2. Explain and implement basic neural network models, including perceptron, ADALINE, MADALINE, and backpropagation networks.
3. Analyze advanced neural network architectures such as Hopfield Networks, Boltzmann Machines, RBF Networks, SOM, ART, and Gaussian Mixture Models.
4. Apply supervised, unsupervised, and reinforcement learning techniques to solve real-world computational problems.
5. Demonstrate foundational knowledge of Natural Language Processing (NLP) including text preprocessing, language models, and parsing techniques.

Learning Outcomes:

At the successful completion of the course, students will be able to:

1. Compare and contrast soft computing techniques (fuzzy systems, neural networks, probabilistic models) with traditional hard computing approaches.
2. Design and train basic neural network models using appropriate activation functions and learning algorithms.
3. Evaluate the performance of advanced neural network models such as RBF, SOM, ART, Hopfield, and Boltzmann Machines for pattern recognition and clustering tasks.
4. Implement text preprocessing and language modelling techniques including N-grams and probabilistic models for NLP applications.
5. Analyse and interpret parsing techniques (constituency, dependency, and probabilistic parsing) for syntactic structure identification in natural language.

Module	Module Content	Duration of Module
I	**Soft Computing & Basic Neural Networks** Introduction of soft computing, soft computing vs. hard computing, various types of soft computing techniques, Fuzzy Computing, Supervised, Unsupervised learning, Reinforcement Learning. Supervised Learning Network – Simple Perceptron Networks, Adaptive Linear Neuron, Multiple Adaptive Linear Neurons, Backpropagation Network, Feedforward model, activation function	15
II	**Advanced Neural Network Models** Hopfield Network, Boltzmann Machine Gaussian Mixture Model, Radial Basis Function (RBF) Network, Kohonen Self-Organizing Map (SOM), Adaptive Resonance Theory (ART) Network	15
III	**Natural Language Processing (NLP)** Text Preprocessing, N-grams and Language Models, Introduction to Parsing in NLP, Constituency Parsing (Phrase Structure Parsing), Dependency parsing, Probabilistic Parsing	15
	Total Lectures	45

Books and References:

Sr.No.	Title	Author/s	Publisher	Edition	Year
1.	Neural Networks, Fuzzy Logic And Genetic Algorithms: Synthesis and Applications	S. Rajasekaran & G. A. Vijayalakshmi Pai	PHI Learning	1st	2003
2.	Speech and Language Processing	Daniel Jurafsky & James H. Martin	Pearson / Prentice Hall	2nd Edition	2014

3.	Principles of Soft Computing	S. N. Sivanandam & S. N. Deepa	Wiley-India	2nd Edition	2007
4.	Artificial Neural Networks: A Practical Course	Ivan Nunes da Silva, Danilo Hernane Spatti, Rogerio A. Flauzino, Luisa H. B. Liboni, Silas F. dos Reis Alves	Springer	1st Edition	2016
5.	Neural Networks and Deep Learning: A Textbook	Charu C. Aggarwal	Springer	1st Edition	2018

<table>
<tr><td colspan="2">Program: B.Sc. – Information Technology (2026 - 27)</td><td colspan="2">Semester: VI</td></tr>
<tr><td colspan="2">Course: Advanced Artificial Intelligence LAB</td><td colspan="2">Course Code: UIAAI354P</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td>Lecture
(Hours per week)</td><td>Credit</td><td>Continuous Assessment (CA)
(Marks -20)</td><td>Semester End Examinations (SEE)
(Marks- 30 in Question Paper)</td></tr>
<tr><td>2</td><td>1</td><td>20</td><td>30</td></tr>
</table>

Learning Objectives:

🎬 To understand the concepts of soft computing techniques including neural networks and fuzzy logic.

🎬 To learn the design and implementation of basic and advanced neural network models for pattern recognition and classification.

🎬 To gain practical experience in training neural networks using learning algorithms such as perceptron learning and backpropagation.

🎬 To explore advanced AI models such as Hopfield Networks and Gaussian Mixture Models for clustering and associative memory.

🎬 To develop practical skills in Natural Language Processing (NLP) including text preprocessing, N-gram models, and parsing techniques.

Course Outcomes:

🎬 Students will be able to implement basic neural network models such as Perceptron, ADALINE, and Multi-Layer Perceptron.

🎬 Students will be able to design and apply fuzzy logic systems for solving real-world control problems.

🎬 Students will be able to implement advanced neural network techniques such as Hopfield Networks and Gaussian Mixture Models.

🎬 Students will be able to apply NLP techniques including text preprocessing, N-gram modeling, and syntactic parsing.

🎬 Students will be able to analyze and evaluate the performance of different AI models for classification, clustering, and language processing tasks.

Outline of Syllabus:

Topics		Duration (Lecture)
Module I	**Soft Computing & Basic Neural Networks** **Practical 1: Implementation of Simple Perceptron** • Implement binary classification using Perceptron. • Train on AND/OR logic gates dataset. • Plot decision boundary. **Practical 2: ADALINE and MADALINE Network** • Implement Adaptive Linear Neuron (ADALINE). • Train using gradient descent. • Compare Perceptron vs ADALINE performance. **Practical 3: Backpropagation Network (Feedforward Neural Network)** • Implement Multi-Layer Perceptron (MLP). • Train using backpropagation algorithm. • Experiment with different activation functions (Sigmoid, ReLU, Tanh). • Analyze learning curves. **Practical 4: Fuzzy Logic System Design** • Design a simple fuzzy inference system. • Define membership functions. • Implement fuzzification, rule evaluation, defuzzification. • Example: Temperature control system.	
Module II	**Advanced Neural Network Models** **Practical 5: Hopfield Network Implementation** • Implement Hopfield Network. • Store and retrieve binary patterns. • Demonstrate associative memory behavior. • Analyze energy convergence. **Practical 6: Gaussian Mixture Model (GMM)** • Implement GMM using Expectation-Maximization (EM) algorithm. • Apply to clustering dataset. • Compare with K-means clustering.	

Module III:	**Natural Language Processing** **Practical 7: Text Preprocessing and N-gram Model** ● Perform tokenization, stop-word removal, stemming/lemmatization. ● Generate Unigram, Bigram, Trigram models. ● Calculate probabilities of word sequences. **Practical 8: NLP Parsing** ● Perform Constituency Parsing using CFG. ● Perform Dependency Parsing using spaCy. ● Implement simple Probabilistic Context-Free Grammar (PCFG). ● Compare parse trees for ambiguous sentences.		
Total Lectures			**30**

Books and References:

Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Neural Networks,A Fuzzy Logic and Genetic Algorithms: Synthesis and Applications	S. Rajasekaran & G. A. Vijayalakshmi Pai	PHI Learning	1st	2003
2.	Speech and Language Processing	Daniel Jurafsky & James H. Martin	Pearson / Prentice Hall	2nd Edition	2014
3.	Principles of Soft Computing	S. N. Sivanandam & S. N. Deepa	Wiley-India	2nd Edition	2007
4.	Artificial Neural Networks: A Practical Course	Ivan Nunes da Silva, Danilo Hernane Spatti, Rogerio A. Flauzino, Luisa H. B. Liboni, Silas F. dos Reis Alves	Springer	1st Edition	2016
5.	Neural Networks and Deep Learning: A Textbook	Charu C. Aggarwal	Springer	1st Edition	2018

<table>
<tr><td colspan="2">Course: Ethical Hacking</td><td colspan="2">Course Code: UIETH355</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td>Lecture
(Hours per week)</td><td>Credit</td><td>Continuous Assessment (CA)
(Marks - 40)</td><td>Semester End Examinations (SEE)
(Marks- 60 in Question Paper)</td></tr>
<tr><td>3</td><td>3</td><td>40</td><td>60</td></tr>
</table>

Learning Objectives:

1. To make the learner understand the significance of Ethical hacking through practice.
2. To make the learner aware of different types of Malware and attacks
3. To give the learner a brief understanding of Social Engineering
4. To provide brief knowledge of hacking webservers and web application

Course Outcomes:

1. Learner will develop an understanding of Ethical hacking and how it can be used to protect different attacks on the internet
2. The learner can Plan a vulnerability assessment and penetration test for a network.
3. The learner can report on strengths and vulnerability of the penetration test performed

Outline of Syllabus:

<table>
<tr><th>Module</th><th>Module Content</th><th>Duration</th></tr>
<tr><td>I</td><td>Introduction to Ethical Hacking:
Overview of Ethics, Overview of Ethical Hacking, Attack Modeling, Methodology of Ethical Hacking

Security Foundations
The Triad, Information Assurance and Risk, Policies, Standards, and Procedures, Organizing Your Protections, Security Technology, Being Prepared

Malware
Malware Types, Malware Analysis, Creating Malware, Malware Infrastructure, Antivirus Solutions, Persistence

Footprinting, Reconnaissance and Social Engineering
Open-Source Intelligence, Domain Name System, Passive Reconnaissance, Website Intelligence, Technology Intelligence
Social Engineering, Physical Social Engineering, Phishing Attacks, Social Engineering for Social Networking, Website Attacks, Wireless Social Engineering, Automating Social Engineering</td><td>15</td></tr>
</table>

II	**Scanning Networks and Enumeration** Ping Sweeps, Port Scanning, Vulnerability Scanning, Packet Crafting and Manipulation, Evasion Techniques, Protecting and Detecting Service Enumeration, Remote Procedure Calls, Server Message Block, Simple Network Management Protocol, Simple Mail Transfer Protocol, Web‑Based Enumeration **System Hacking** Searching for Exploits, System Compromise, Gathering Passwords Password Cracking, Client‑Side Vulnerabilities, Living Off the Land, Fuzzing, Post Exploitation **Sniffing** Packet Capture, Detecting Sniffers, Packet Analysis, Spoofing Attacks **Wireless Security** Wi‑Fi, Bluetooth, Mobile Devices	15
III	**Attack and Defense** Wireless Security, Web Application Attacks, Denial‑of‑Service Attacks, How DDoS Attacks Work, DOS /DDoS Countermeasures, Application Exploitation, Lateral Movement, Defense in Depth/Defense in Breadth, Defensible Network Architecture **SQL Injection:** Finding SQL Injection Vulnerability, Purpose of SQL Injection, SQL Injection Countermeasures **Cryptography:** Basic Encryption, Symmetric Key Cryptography, Asymmetric Key Cryptography, Certificate Authorities and Key Management, Cryptographic Hashing, PGP and S/MIME, Disk and File Encryption **Application of AI:** AI-based tools like **Burp Suite AI Extensions** analyze web traffic patterns to detect **SQL Injection (SQLi), Cross-Site Scripting (XSS), and CSRF attacks**	15
	Total Lectures	45

Textbook(s):

1. CEH v12 Certified Ethical Hacker Study Guide, Ric Messier, 2023
2. Certified Ethical Hacker Study Guide, Kimberly Graves, 2010
3. ALL IN ONE CEH Certified Ethical Hacker Exam Guide, Third Edition, Matt Walker, 2017

Additional Reference(s):

1. Certified Ethical Hacker: Michael Gregg, Pearson Education,1st Edition, 2013
2. Certified Ethical Hacker: Matt Walker, TMH, 2011
3. Ethical Hacking Review Guide Kimberly Graves Wiley Publishing
4. Ethical Hacking Ankit Fadia 2nd Edition Macmillan India Ltd, 2006

Program: B.Sc. – Information Technology (2026 - 27)		Semester: VI	
Course: Ethical Hacking LAB		**Course Code: UIETH355P**	
Teaching Scheme		**Evaluation Scheme**	
Lecture (Hours per week)	**Credit**	**Continuous Assessment (CA) (Marks - 20)**	**Semester End Examinations (SEE) (Marks- 30 in Question Paper)**
2	1	20	30

Learning Objectives:

🎬 To understand the fundamental concepts of ethical hacking and cybersecurity practices.

🎬 To learn the installation and configuration of security testing environments such as Kali Linux.

🎬 To gain knowledge of information gathering and network scanning techniques used in security assessment.

🎬 To understand vulnerability assessment and exploitation techniques using ethical hacking tools.

🎬 To develop practical skills in identifying and mitigating security threats in computer systems and web applications.

Course Outcomes:

🎬 Students will be able to set up and use penetration testing environments using Kali Linux and basic Linux commands.

🎬 Students will be able to perform information gathering and network scanning using tools like Google Hacking and Nmap.

🎬 Students will be able to conduct vulnerability assessments using tools such as Nessus and OpenVAS.

🎬 Students will be able to demonstrate basic exploitation techniques using tools like Metasploit and Burp Suite.

🎬 Students will be able to analyze and test system security by performing password cracking and identifying vulnerabilities.

Outline of Syllabus:

	List of Practicals :	Duration (Lecture)

	1. Installations and Foundation • Install and configure Kali Linux on Virtual Machine • Basic Linux Commands 2. Information gathering • Use google hacking to gather information about target • Practice OSINT to collect information about potential targets 3. Conduct a port and network scan using “Nmap” to discover devices and open ports 4. Perform banner grabbing using “Netcat” 5. Conduct vulnerability scanning through “Nessus” and “OpenVAS” 6. Run a simulated malware attack using a simple virus 7. Demonstrate privilege escalation techniques using tools like “sudo” or “su” 8. Exploitation • Exploit a vulnerable web application using “SQL injection” • Use “Metasploit” to demonstrate exploit techniques 9. Use “Burp Suite” for intercepting and manipulating web requests 10. Crack password using “hashcat” and “hydra”	
Total Lectures		**30**

<table>
<tr><td colspan="3">Program: B.Sc. – Information Technology (2026 - 27)</td><td>Semester: VI</td></tr>
<tr><td colspan="3">Course: Search Engine Optimization</td><td>Course Code: UISEO356</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td>Lecture
(Hours per week)</td><td>Credit</td><td>Continuous Assessment (CA)

(Marks - 20)</td><td>Semester End Examinations (SEE)
Marks - 30 in Question Paper)</td></tr>
<tr><td>2</td><td>2</td><td>20</td><td>30</td></tr>
</table>

Learning Objectives:

1. Understand the core principles of search engines, including their functionality, crawling, indexing, and ranking mechanisms
2. Apply keyword research methods and SEO-friendly web development techniques to design and build optimized websites
3. Analyze website performance using SEO analytics tools and recommend strategies for improvement based on data insights

Course Outcomes:

After completion of the course, learners would be able to:

CO1: Describe how search engines work and explain the factors that influence webpage ranking CO2: Use technical tools for keyword research and develop websites that are optimized for search engines, considering accessibility, content structuring, and URL hierarchy.

CO3: Diagnose SEO issues using tools like Google Search Console, identify technical bottlenecks (e.g., slow server responses, broken links), and implement solutions for improved site performance

Outline of Syllabus: (per session plan)

<table>
<tr><th>Module</th><th>Description</th><th>No of Hours</th></tr>
<tr><td>1</td><td>Introduction to Search Engines and their Fundamentals

Chapter 1. Putting Search Engines in Context and Meeting Them
Identifying Search Engine Users, Understanding the Search Engines: They’re a Community, Finding the Common Threads among the Engines, Getting to Know the Major Engines, Understanding Metasearch Engines

Chapter 2. Search Fundamentals
Deconstructing Search, The Language of Search, Crawling, The Index, The Search Engine Results Page, Ranking Factors</td><td>08</td></tr>
<tr><td>2</td><td>Keyword Research and SEO-Friendly Web Design

Chapter 1. Keyword Strategy
Researching Client Niche Keywords, Checking Out Seasonal Keyword Trends, The Words and Phrases That Define Business, Internal Resources for Keyword Research, External Resources for Keyword Research, Keyword Valuation, Acting on Keyword Plan, Periodic Keyword Reviews

Chapter 2. Developing SEO Friendly Website
Making Site Accessible to Search Engines, Creating Optimal Information Architecture, Root Domains, Subdomains, and Microsites, Optimization of Domain Names/URLs, Keyword</td><td>12</td></tr>
</table>

	Targeting, Content Optimization and Duplicate Issues, Controlling Content with Cookies and Session IDs, Content Delivery and Search Spider Control, Best Practices for Multilingual/Multicountry Targeting, Google's EEAT and YMYL, Domain Changes, Content Moves, and Redesigns	
3	**SEO Optimization and Analytics** Chapter 1. Optimizing the Foundations Meeting the Servers, Health and Fast Servers, Excluding Pages and Sites from the Search Engines, Creating 404 Error Pages, Dirty IPs , Serving Your Site to Different Devices, Selecting Domain Name, Discovering the Types of Redirects, Reconciling www and Non-www URLs, 301 Redirects, Inviting Spiders to Your Site, 302 Hijacks, Handling Secure Server Problems Chapter 2. SEO Analytics and Measurement Why Measurement Is Essential in SEO, Analytics Tools for Measuring Search Traffic, Connecting SEO and Conversions, Diagnostic Search Metrics, Specific Analytics Tools from Google and Bing, Auditing Websites for SEO Improvements, AI in Search Engine **AI Content Writing** – Tools like **ChatGPT, Jasper AI, and Copy.ai generate SEO-friendly content with relevant keywords**	10
Total		**30**

Important Note:

1. Students must implement a basic SEO on a small website or a blog
2. Compare high-competition keywords vs. long-tail keywords for a hypothetical ecommerce store
3. Present scenarios of poorly optimized websites and how to address issues like duplicate content, bad URL structures, or missing meta tags. For example: Audit a website with broken links and identify the impact on user experience and ranking.
4. Discuss cases of unethical SEO practices, such as keyword stuffing, link schemes, and cloaking, and their consequences.
5. Discuss key algorithm updates like Panda, Penguin, and Hummingbird to highlight how search engines evolve and affect SEO practices
6. Discuss scenarios where websites failed to adapt to mobile-first indexing or voice search trends. For example: Compare desktop vs. mobile performance of a website

Reference Books:

[1] Eric Enge, Stephen Spencer, Jessie Stricchiola, "The Art of SEO", 4th Edition, O'Reilly Media Inc, September 2023

[2] Bruce Clay, Kristopher B. Jones, "Search Engine Optimization All-in-One For Dummies", 4th Edition, For Dummies, February 2022 (9 Books in One)

Program: B.Sc. – Information Technology (2026 - 27)	**Semester: VI**
Course: Hybrid Mobile Application LAB	**Course Code: UIHMA358P**

Teaching Scheme		Evaluation Scheme	
Lecture (Hours per week)	Credit	Continuous Assessment (CA) (Marks - 20)	Semester End Examinations (SEE) (Marks- 30 in Question Paper)
2	1	20	30

Learning Objectives:

🎬 To understand the fundamentals of hybrid mobile application development using Flutter and Dart.

🎬 To learn the basics of Dart programming including functions, control structures, and user input handling.

🎬 To gain practical experience in designing user interfaces using Flutter widgets, layouts, and navigation.

🎬 To develop mobile applications with form handling, validation, and user interaction components.

🎬 To integrate backend services such as Firebase and implement features like notifications, themes, and progress indicators.

Course Outcomes:

🎬 Students will be able to develop basic mobile applications using Flutter and Dart.

🎬 Students will be able to design responsive user interfaces using Flutter widgets, layouts, and navigation components.

🎬 Students will be able to implement application logic using Dart functions and programming constructs.

🎬 Students will be able to create interactive applications using forms, validation, toast messages, and themes.

🎬 Students will be able to integrate Firebase services and implement advanced UI elements such as progress indicators in mobile applications.

Outline of Syllabus:

	List of Practicals :	Duration (Lecture)
	1. Create a Hello World Application in Flutter.	

	2. Write an application as ask the user for a number and determine whether the number is prime or not in Dart. 3. Write a program in Dart that find the area of a circle using function. 4. Layouts: Create a simple UI with rows and columns of coloured containers using Flutter’s widget-based architecture 5. Create a simple Calculator application in Flutter. 6. Create a simple Navigation App. 7. Designing a mobile app to calculate the income tax of the user 8. Create forms in Flutter, Implementing Text input, drop-down lists, and more form elements in forms, Applying validation in form. 9. Build an application for adding firebase to Flutter application. 10. Construct mobile applications to display Toast message using Flutter 11. Write a program to implement Themes and Styles 12. Design a mobile app implementing Progress Indicators	
Total Lectures		**30**

Program: B.Sc. - Information Technology (2026 - 27)	Semester: II
Course: On Job Training	**Course Code:** UIOJT359

On-Job Training (OJT) Component for B.Sc.-IT

A. Introduction

The B.Sc. IT program at SVKM's Usha Pravin Gandhi College of Arts, Science and Commerce incorporates a comprehensive On-Job Training (OJT) module, designed to bridge the gap between academic learning and real-world industry practices. This mandatory component is structured to enrich our students' educational experience, providing them with hands-on exposure to practical work scenarios and enhancing their employability and skill set.

- **Importance of OJT:**

  This direct engagement with the industry and research projects not only enhances students' learning experiences but also significantly boosts their confidence, problem-solving abilities, and readiness for employment.

- **Scope of OJT:**

  OJT is an integral part of the curriculum, scheduled during the semester, offering students the opportunity to work on actual projects within industries, research institutions, and other professional environments. This experiential learning phase allows students to apply theoretical knowledge to practical challenges, fostering a deeper understanding of their field.

B. Conduct of OJT:

- **Mode of Participation:** Students have the flexibility to undertake their OJT in a physical mode within the industry or online, depending on the nature of the project and the hosting organization's policies.
- **On-Premise for Research Projects:** For those inclined towards research, OJT can be conducted on-premise, assisting in ongoing research projects at research institutions, offering a unique insight into the research methodologies and challenges.

C. On-the-Job Training for Industry Readiness

- Hands-On Experience: B.Sc. IT students will engage in a 4-6 week OJT program, accumulating 120 hours of practical experience. This immersive learning opportunity allows students to apply theoretical knowledge in real-world settings.
- Diverse Job Opportunities: During OJT, students can explore various roles such as software developer, system analyst, database administrator, cybersecurity analyst, IT consultant, and more. These roles align with their academic background and provide valuable industry exposure.
- Industry-Relevant Skills: OJT enables students to develop technical skills in programming languages, software development methodologies, database management, network security,

and other IT domains. Additionally, they gain soft skills like communication, teamwork, problem-solving, and time management.
- External Placement: Students are responsible for securing their OJT placements however institution will also provide guidance and support from for internships in reputable organizations/startups etc., ensuring exposure to diverse work environments and cutting-edge technologies.
- Career Readiness: By bridging the gap between academia and industry, OJT prepares B.Sc. IT students for successful careers in fields such as artificial intelligence, data science, cloud computing, cybersecurity, and software engineering. This practical experience enhances their employability and professional competency.

D. OJT Opportunities Across Various Organizations

Internship Placements: Students are encouraged to seek OJT experiences in a wide range of organizations, including:

- Software Development Firms: Students can immerse themselves in software development and programming tasks, gaining practical skills and insights into industry best practices.
- Hardware/Manufacturing Companies: Opportunities abound for students to delve into hardware design, manufacturing processes, and quality assurance protocols, providing valuable hands-on experience in technological innovation.
- Small-Scale Industries/Service Providers: Students have the chance to explore diverse sectors such as banking, healthcare clinics, NGOs, and professional institutions like CA firms or law firms. This exposure allows for a comprehensive understanding of industry operations and service delivery.
- Civic Departments: Engagement with local civic departments such as ward offices, post offices, police stations, or panchayats offers students a unique opportunity to observe governance structures and contribute to community activities, fostering civic responsibility and practical knowledge.
- Research Centers/University Departments/Colleges: Students can actively participate as research assistants or in similar capacities within research projects or initiatives. This collaboration between academia and industry fosters innovation and enriches academic learning with real-world applications.

E. Evaluation:

As part of our internship program, students need to submit two important documents:

Daily Diary:

- Students must write daily entries in an online diary, including what they did each day during the internship.
- Each entry should be short, just 3-4 sentences, summarizing the day's activities.
- Faculty members will regularly check these entries to monitor progress.

OJT Report:

It includes:

- A Certificate: This is a paper from the organization where the student did the internship, saying they completed it.
- Title: This explains what the student did during the internship.
- Information about the organization: A short description of where the student worked.
- Job Description: A summary of the tasks and activities the student was given.
- Self-assessment: The student's own thoughts on what they learned during the internship.

The OJT is evaluated on a dual-component basis to ensure a comprehensive assessment of the student's performance and learning outcomes.

- Supervisor's Feedback: Each student will have a supervisor in the organization where they intern. The supervisor will give feedback on how well the student did their tasks, solved problems, and behaved professionally.
- Institutional Review: After finishing the internship, students will give a presentation at the college. They'll talk about what they did during the internship and how it helped them learn. Teachers will watch and evaluate the presentation.

F. OJT workload for the faculty

Each student is paired with a faculty mentor who guides them through their OJT experience. The faculty mentor oversees a group of students, serving as their main point of contact throughout the process. They monitor the progress of each student's OJT by reviewing their diary entries, coordinating with industry mentors, and offering support with report writing. To ensure effective mentorship, faculty members are entitled to one hour of workload allocation per week for their mentoring responsibilities.

<table>
<tr><td colspan="2">Program: B.Sc. – Information Technology (2026 - 27)</td><td colspan="2">Semester: VI</td></tr>
<tr><td colspan="2">Course: Project</td><td colspan="2">Course Code: UIPRJ357</td></tr>
<tr><td colspan="2">Teaching Scheme</td><td colspan="2">Evaluation Scheme</td></tr>
<tr><td>Lecture
(Hours per week)</td><td>Credit</td><td>Continuous Assessment (CA)
(Marks - 20)</td><td>Semester End Examinations (SEE)
(Marks- 30 in Question Paper)</td></tr>
<tr><td>2</td><td>1</td><td>20</td><td>30</td></tr>
</table>

Project Dissertation & Implementation Semester VI

Chapter 1 to 4 should be submitted in Semester V in spiral binding. These chapter have also to be included in Semester VI report. Semester VI report must be hard bound with golden embossing. Students will be evaluated based on the dissertation in semester V and dissertation and viva voce in Semester VI.

Types of the Project

Most of the students are expected to work on a real-life project preferably in some industry/ Research and Development Laboratories/Educational Institution/Software Company. Students are encouraged to work in the various software application domains. However, it is ***not mandatory*** for a student to work on a real-life project. The student can formulate a project problem with the help of her/his Guide and submit the project proposal of the same. **Approval of the project proposal is mandatory**. If approved, the student can commence working on it, and complete it. Use the latest versions of the software packages for the development of the project.

The project report should be documented with scientific approach to the solution of the problem that the students have sought to address. The project report should be prepared to solve the problem in a methodical and professional manner, making due references to appropriate techniques, technologies, and professional standards. The student should start the documentation process from the first phase of software development so that one can easily identify the issues to be focused upon in the ultimate project report. The student should also include the details from the project diary, in which they will record the progress of their project throughout the course. The project report should contain enough details to enable examiners to evaluate the work. The important points should be highlighted in the body of the report, with details often referred to appendices.

Format of the Project Report

Title Page

Original Copy of the Approved Proforma of the Project

Proposal Certificate of Authenticated work

Role and

Responsibility Form

Abstract

Acknowledgement

Table of Contents Table of Figures

CHAPTER 1: INTRODUCTION

1.1 Background

1.2 Objectives

1.3 Purpose, Scope, and Applicability

1.3.1 Purpose

1.3.2 Scope

1.3.3 Applicability

1.4 Achievements

1.5 Organization of Report

CHAPTER 2: SURVEY OF

TECHNOLOGIES

CHAPTER 3: REQUIREMENTS AND ANALYSIS

3.1 Problem Definition

3.2 Requirements Specification

3.3 Planning and Scheduling

3.4 Software and Hardware Requirements

3.5 Preliminary Product Description

3.6 Conceptual Models

CHAPTER 4: SYSTEM DESIGN

4.1 Basic Modules

4.2 Data Design

4.2.1 Schema Design

Explanation of the Content

Title Page

Sample format of Title page is given in Appendix 1 of this block. Students should follow the given format.

Original Copy of the Approved Proforma of the Project Proposal

Sample Proforma of Project Proposal is given in Appendix 2 of this block. Students should follow the given format.

Certificate of Authenticated work

Sample format of Certificate of Authenticated work is given in Appendix 3 of this block. Students should follow the given format.

Role and Responsibility Form

Sample format for Role and Responsibility Form is given in Appendix 4 of this block. Students should follow the given format.

Abstract

This should be one/two short paragraphs (100-150 words total), summarizing the project work. It is important that this is not just a re-statement of the original project outline. A suggested flow is background, project aims and main achievements. From the abstract, a reader should be able to ascertain if the project is of interest to them and, it should present results of which they may wish to know more details.

Acknowledgements

This should express student's gratitude to those who have helped in the preparation of project.

Table of Contents: The table of contents gives the readers a view of the detailed structure of the report. The students would need to provide section and subsection headings with associated pages. The formatting details of these sections and subsections are given below.

Table of Figures: List of all Figures, Tables, Graphs, Charts etc. along with their page numbers in a table of figures.

Chapter 1: Introduction

The introduction has several parts as given below:

Background: A description of the background and context of the project and its relation to work already done in the area. Summaries existing work in the area concerned with the project work.

Objectives: Concise statement of the aims and objectives of the project. Define exactly what is going to be done in the project; the objectives should be about 30 /40 words.

Purpose, Scope, and Applicability: The description of Purpose, Scope, and Applicability are given below:

- Purpose: Description of the topic of the project that answers questions on why this project is being done. How the project could improve the system its significance and theoretical framework.
- Scope: A brief overview of the methodology, assumptions, and limitations. The students should answer the question: What are the main issues being covered in the project? What are the main functions of the project?
- Applicability: The student should explain the direct and indirect applications of their work. Briefly discuss how this project will serve the computer world and people.

Achievements: Explain what knowledge the student achieved after the completion of the work. What contributions has the project made to the chosen area? Goals achieved - describes the degree to which the findings support the original objectives laid out by the project. The goals may be partially or fully achieved, or exceeded.

Organization of Report: Summarizing the remaining chapters of the project report, in effect, giving the reader an overview of what is to come in the project report.

Chapter 2: Survey of Technologies

In this chapter Survey of Technologies should demonstrate the student’s awareness and understanding of Available Technologies related to the topic of the project. The student should give the detail of all the related technologies that are necessary to complete the project. They should describe the technologies available in the chosen area and present a comparative study of all those Available Technologies. Explain why the student selected the one technology for the completion of the objectives of the project.

Chapter 3: Requirements and Analysis

Problem Definition: Define the problem on which the students are working in the project. Provide details of the overall problem and then divide the problem in to sub-problems. Define each sub-problem clearly.

Requirements Specification: In this phase the student should define the requirements of the system, independent of how these requirements will be accomplished. The Requirements Specification describes the things in the system and the actions that can be done on these things. Identify the operation and problems of the existing system.

Planning and Scheduling: Planning and scheduling is a complicated part of software development. Planning, for our purposes, can be thought of as determining all the small tasks that must be carried out to accomplish the goal. Planning also considers, rules, known as constraints, which, control when certain tasks can or cannot happen. Scheduling can be thought of as determining whether adequate resources are available to carry out the plan. The student should show the Gantt chart and Program Evaluation Review Technique (PERT).

Software and Hardware Requirements: Define the details of all the software and hardware needed for the development and implementation of the project.

- Hardware Requirement: In this section, the equipment, graphics card, numeric co-processor, mouse, disk capacity, RAM capacity etc. necessary to run the software must be noted.
- Software Requirements: In this section, the operating system, the compiler, testing tools, linker, and the libraries etc. necessary to compile, link and install the software must be listed. Preliminary Product Description: Identify the requirements and objectives of the new system. Define the functions and operation of the application/system the students are developing as project.

Conceptual Models: The student should understand the problem domain and produce a model of the system, which describes operations that can be performed on the system, and the allowable sequences of those operations. Conceptual Models could consist of complete Data Flow Diagrams, ER diagrams, Object-oriented diagrams, System Flowcharts etc.

Chapter 4: System Design

Describes desired features and operations in detail, including screen layouts, business rules, process diagrams, pseudocode, and other documentation.

Basic Modules: The students should follow the divide and conquer theory, so divide the overall problem into more manageable parts and develop each part or module separately. When all modules are ready, the student should integrate all the modules into one system. In this phase, the student should briefly describe all the modules and the functionality of these modules.

Data Design: Data design will consist of how data is organized, managed, and manipulated.

- Schema Design: Define the structure and explanation of schemas used in the project.
- Data Integrity and Constraints: Define and explain all the validity checks and constraints provided to maintain data integrity.

Procedural Design: Procedural design is a systematic way for developing algorithms or procedurals.

- Logic Diagrams: Define the systematical flow of procedure that improves its comprehension and helps the programmer during implementation. e.g., Control Flow Chart, Process Diagrams etc.
- Data Structures: Create and define the data structure used in procedures.
- Algorithms Design: With proper explanations of input data, output data, logic of processes, design and explain the working of algorithms.

User Interface Design: Define user, task, environment analysis and how to map those requirements to develop a “User Interface.” Describe the external and internal components and the architecture of user interface. Show some rough pictorial views of the user interface and its components.

Security Issues: Discuss Real-time considerations and Security issues related to the project and explain how the student intends avoiding those security problems. What is the security policy plans and architecture?

Test Cases Design: Define test cases, which will provide easy detection of errors and mistakes with in a minimum period and with the least effort. Explain the different conditions in which the students wish to ensure the correct working of the project.

Chapter 5: Implementation and Testing

Implementation Approaches: Define the plan of implementation, and the standards the students have used in the implementation.

Coding Details and Code Efficiency: Students not need include full source code, instead, include only the important codes (algorithms, applets code, forms code etc.). The program code should contain comments needed for explaining the work a piece of code does. Comments may be needed to explain why it does it, or, why it does a particular way.

The student can explain the function of the code with a shot of the output screen of that program code.

Code Efficiency: The student should explain how the code is efficient and how the students have handled code optimization.

Testing Approach: Testing should be according to the scheme presented in the system design chapter and should follow some suitable model – e.g., category partition, state machine-based. Both functional testing and user-acceptance testing are appropriate. Explain the approach of testing.

- Unit Testing: Unit testing deals with testing a unit or module. This would test the interaction of many functions but, do confine the test within one module.
- Integrated Testing: Brings all the modules together into a special testing environment, then checks for errors, bugs, and interoperability. It deals with tests for the entire application. Application limits and features are tested here.

Modifications and Improvements: Once the students finish the testing they are bound to be faced with bugs, errors and they will need to modify your source code to improve the system. Define what modification are implemented in the system and how it improved the system.

Chapter 6: Results and Discussion

Test Reports: Explain the test results and reports based on the test cases, which should show that the project can face any problematic situation and that it works fine in different conditions. Take the different sample inputs and show the outputs.

User Documentation: Define the working of the software; explain its different functions, components with screen shots. The user document should provide all the details of the product in such a way that any user reading the manual, is able to understand the working and functionality of the document.

Chapter 7: Conclusions

Conclusion: The conclusions can be summarized in a short chapter (2 or 3 pages). This chapter brings together many of the points that would have made in the other chapters.

Limitations of the System: Explain the limitations encountered during the testing of the project that the students were not able to modify. List the criticisms accepted during the demonstrations of the project.

Future Scope of the Project describes two things: firstly, new areas of investigation prompted by developments in this project, and secondly, parts of the current work that was not completed due to time constraints and/or problems encountered.

REFERENCES

It is very important that the students acknowledge the work of others that they have used or adapted in their own work, or that provides the essential background or context to the project. The use of references is the standard way to do this. Please follow the given standard for the references for books, journals, and online material. The citation is mandatory in both the reports.

E.g.: Linhares, A., & Brum, P. (2007). Understanding our understanding of strategic scenarios: What role do chunks play? *Cognitive Science*, *31*(6), 989-1007. https://doi.org/doi:10.1080/03640210701703725

Lipson, Charles (2011). Cite right: A quick guide to citation styles; MLA, APA, Chicago, the sciences, professions, and more (2nd ed.). Chicago [i.e.]: University of Chicago Press. p. 187. ISBN 9780226484648.

Elaine Ritchie, J Knite. (2001). *Artificial Intelligence, Chapter 2 ,p.p 23 - 44.* Tata McGraw-Hill.

GLOSSARY

If you the students any acronyms, abbreviations, symbols, or uncommon terms in the project report then their meaning should be explained where they first occur. If they go on to use any of them extensively then it is helpful to list them in this section and define the meaning.

APPENDICES

These may be provided to include further details of results, mathematical derivations, certain illustrative parts of the program code (e.g., class interfaces), user documentation etc. If there are technical details of the work done that might be useful to others who wish to build on this work, but that are not sufficiently important to the project to justify being discussed in the main body of the project, then they should be included as appendices.

SUMMARY

Project development usually involves an engineering approach to the design and development of a software system that fulfils a practical need. Projects also often form an important focus for discussion at interviews with future employers as they provide a detailed example of what the students can achieve. In this course the students can choose your project topic from the lists given in Unit 4: Category-wise Problem

Definition.

FURTHER READINGS

1. Modern Systems Analysis and Design; Jeffrey A. Hoffer, Joey F. George, Joseph,S. Valacich; Pearson Education; Third Edition; 2002.

2. ISO/IEC 12207: Software Life Cycle Process (http://www.software.org/quagmire/descriptions/iso-iec12 207.asp).

3. IEEE 1063: Software User Documentation (http://ieeexplore.ieee.org).

4. ISO/IEC: 18019: Guidelines for the Design and Preparation of User Documentation for Application Software.

5. http://www.sce.carleton.ca/squall.

6. http://en.tldp.org/HOWTO/Software-Release-Practice-HOWTO/documentation.html.

7. http://www.sei.cmu.edu/cmm/

PROFORMA FOR THE APPROVAL PROJECT PROPOSAL

(Note: All entries of the proforma of approval should be filled up with

appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)

PNR **No.: ……………………** Roll no**:** ___________

1. Name of the Student ____

2. Title of the Project

   ____

3. Name of the Guide

   ____

4. Teaching experience of the Guide ________________________________

5. Is this your first submission? Yes ☐ No ☐

Signature of the Student Signature of the Guide

Date: ………………… Date: ……………………

Signature of the

Coordinator Date:

…………………

(All the text in the report should be in times new roman)

TITLE OF THE PROJECT (NOT EXCEEDING 2 LINES, 24 BOLD, ALL CAPS)

A Project Report (12 Bold)

Submitted in partial fulfillment of the Requirements for the award of the Degree of (size-12)

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY) (14 BOLD, CAPS)

By(12 Bold)

Name of The Student (size-15, title case) Seat Number (size-15)

Under the esteemed guidance of (13 bold)
Mr./Mrs. Name of The Guide (15 bold, title case)
Designation (14 Bold, title case)

COLLEGE LOGO

DEPARTMENT OF INFORMATION TECHNOLOGY(12 BOLD, CAPS)
COLLEGE NAME (14 BOLD, CAPS)

(Affiliated to University of Mumbai) (12, Title case, bold, italic) **CITY, PIN CODE(12 bold, CAPS) MAHARASHTRA (12 bold, CAPS)**
YEAR (12 bold)

COLLEGE NAME (14 BOLD, CAPS)

(Affiliated to University of Mumbai) (13, bold, italic)

CITY-MAHARASHTRA-PINCODE(13 bold, CAPS)

DEPARTMENT OF INFORMATION TECHNOLOGY (14 BOLD, CAPS)

College Logo

CERTIFICATE (14 BOLD, CAPS, underlined, centered)

This is to certify that the project entitled, **"Title of The Project "**, is bonafied work of **NAME OF THE STUDENT** bearing Seat. No: (**NUMBER**) submitted in partial fulfillment of the requirements for the award of degree of BACHELOR OF SCIENCE in INFORMATION TECHNOLOGY from University of Mumbai. (12, times new roman, justified)

Internal Guide (12 bold) **Coordinator**

(Don"t write names of lecturers or HOD)

External Examiner

Date: **College Seal**

COMPANY CERTIFICATE (if applicable)

(Project Abstract page format)

Abstract (20bold, caps, centered)

Content (12, justified)

Note: Entire document should be with 1.5 line spacing and all paragraphs should start with 1 tab space.

ACKNOWLEDGEMENT
(20, BOLD, ALL CAPS, CENTERED)

The acknowledgement should be in times new roman, 12 fonts with 1.5 line spacing, justified.

(Declaration page format)

DECLARATION (20 bold, centered, all caps)

Content (12, justified)

I hereby declare that the project entitled, "**Title of the Project**" done at **place where the project is done**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfillment of the requirements for the award of degree of **BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)** tube submitted as final semester project as part of our curriculum.

Name and Signature of the Student

TABLE OF CONTENTS (20bold, caps, centered)

Should be generated automatically using word processing software.

Chapter 7: References

List of Tables (20 bold, centered, Title Case)

Should be generated automatically using word processing software.

List of Figures (20 bold, centered, Title Case)

Should be generated automatically using word processing software.

(Project Introduction page format)

Chapter 1

Introduction (20 Bold, centered)

Content or text (12, justified)

Note: Introduction must cover brief description of the project with minimum 4 pages.

Chapter 2

System Analysis (20 bold, Centered)

Subheadings are as shown below with following format (16 bold, CAPS)

2.1 Existing System (16 Bold)

2.1.1 (14 bold, title case)

2.1.1.1 (12 bold, title case)

2.2 Proposed System

2.3 Requirement Analysis

2.4 Hardware Requirements

2.5 Software Requirements

2.6 Justification of Platform – (how h/w & s/w satisfying the project)

Table 2.1: Caption

Chapter 3

System Design (20 bold, centered)

Subheadings are as shown below with following format (16 bold, CAPS)
Specify figures as Fig 11.1 – caption

3.1 Module Division

3.2 Data Dictionary

3.3 E-R Diagrams

3.4 Data Flow Diagrams / UML

Note: write brief description at the bottom of all diagrams

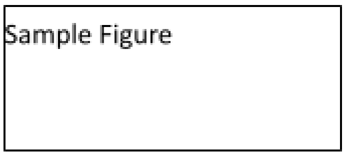

Fig. 3.1: Caption

Chapter 4

Implementation and Testing (20 bold, centered)

4.1 Code (Place Core segments)

Content includes description about coding phase in your project (Font-12)
(* do not include complete code just description)

4.2 Testing Approach

Subheadings are as shown below with following format (16 bold, CAPS)

4.2.1 Unit Testing

4.2.2 Integration Testing

Note:

- Explain about above testing methods
- Explain how the above techniques are applied in your project

Provide Test plans, test cases, etc. relevant to your project

Chapter 5

Results and Discussions(20 bold, centered)

Note: Place Screen Shots and write the functionality of each screen at the bottom

Chapter 6

Conclusion and Future Work (20 bold, centered)

The conclusions can be summarized in a short chapter around 300 words. Also include limitations of your system and future scope (12, justified)

Chapter 7

References (20 bold, centered)

Content (12, LEFT)

[1] Title of the book, Author

[2] Full URL of

~~- online re~~ferences

*** <u>NOTE ABOUT PROJECT VIVA VOCE:</u>**

Student may be asked to write code for problem during VIVA to demonstrate his coding capabilities and he/she may be asked to write any segment of coding used in the in the project. The project can be done in group of at most four students. However, the length and depth of the project should be justified for the projects done in group. A big project can be modularized and different modules can be assigned as separate project to different students.

Marks Distribution:

Semester V: 50 Marks

Documentation: 50 marks

Semester VI: 150 Marks

Documentation: 50 Marks:

Implementation and Viva Voce: 100 Marks

The plagiarism should be maintained as per the UGC guidelines.

Evaluation Pattern

Examination pattern from AY 26-27 onwards

Credits assigned	Max. marks allocated to a course	Total max. marks for Semester End Examination	Total max. marks for Continuous Assessment
03 / 04 credit course	100 marks	60 marks (Exam duration – Two Hours)	# 40 marks
01 / 02 credit course	50 marks	30 marks (Exam duration – One Hour)	## 20 marks:

<u># Three sub components –</u>

i) **10 marks + 10marks** (three tests* for 10 marks each to be conducted at different instants of time amongst which best two out of three will be considered)

ii) **20 marks** assignments / projects / presentations etc.

<u>## Two sub components-</u>

iii) **10 marks** (two tests* of 10 marks each to be conducted at different instants of time and average of two will be considered)

iv) **10 marks** assignments / projects / presentations etc.

*Class tests / Critical Essays / Article Reviews / Open Book Tests / Quizzes etc.
Duration for examination: 60 marks- 2 hours, 30 marks -1-hour, 10 marks-20 Minutes

External Component Paper Pattern (TY BSC IT Major 60 Marks)

Que 1	(Any 4 out of 6) 05 x 4 (M1)	(20)
Que 2	(Any 4 out of 6) 05 x 4 (M2)	(20)
Que 3	(Any 4 out of 6) 05 x 4 (M3)	(20)

External Component Paper Pattern (TY BSC IT Minor, Vocational (if applicable) 30 Marks)

Question No.	Description	Marks	Total marks
Que 1	Answer the following Questions Any 2/3 (M1)	05x2	10
Que 2	Answer the following Questions: (CLO 2) Any 2/3 (M2)	05x2	10

Que 3	Answer the following Questions: (CLO 3) Any 2/3 (M3)	05x2	10
		Total Marks	**30**

Internal Component (Class Test of 10 Marks) Paper Pattern (TY BSC IT Major, Minor, Vocational if applicable)

Que 1	Fill ups / Short Answers / MCQs	(04)
Que 2	Brief Answer 02 x 3 or 03 x 2	(06)

Internal Component [ICA] (TY BSC IT Vocational Practical 20 Marks)

1.	Performance during the regular sessions	15
2.	Viva Voce	5

Rubrics for Practical					
	Performance Indicator	**Poor**	**Average**	**Good**	**Excellent**
1	**Knowledge (Factual /Conceptual/Procedural/cognitive)**	**1**	**2**	**3**	**4**
2	**Describe and Demonstration (Factual /Conceptual/procedural/cognitive)**	**1**	**2**	**3**	**4**
3	**Strategy (Analyze &/or Evaluate) (Factual /Conceptual/procedural/cognitive)**	**1**	**2**	**3**	**4**
4	**Interpret/Develop/ Attitude towards Learning (Factual /Conceptual/procedural/cognitive/ receiving, attending, responding, valuing, organizing, Characterization by value)**	**1**	**2**	**3**	**4**
5	**Nonverbal communication skills/Behavior skills (motor skills, hand-eye coordination, speech behavior)**	**1**	**2**	**3**	**4**

Practical Examination Computer Based Test Pattern

(TY BSC IT Vocational Practical 30 Marks)

Que 1	Question 1 (20 marks) or Que 1a. and Que 1b. (10 marks each)	(20)
Que 2	Journal	(05)
Que 3	Viva	(05)